Cosmology And Black Holes

Leonard Susskind

Supplementary track, 2011 Winter Topics In String Theory
Transcript-derived notes curated by LazyingArt LLC

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC.

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Chapter 1

Lecture 1: Topics in String Theory

These notes follow Leonard Susskind's first lecture in the supplementary track on cosmology and black holes. The lecture begins with a philosophical challenge to reductionism and then steadily sharpens that challenge into a sequence of mathematical examples: bosonization in $1+1$ dimensions, electric–magnetic duality in quantum electrodynamics, the appearance of D-branes in string theory, strong-coupling transitions, emergent geometry, and finally the combinatorial landscape of string vacua. The aim here is to preserve the lecture's order, rhythm, and motivation while giving the mathematics a cleaner written form. Based on a lecture by Leonard Susskind; notes curated by LazyingArt LLC.

1.1 Reductionism, Complexity, and the Question of Simplicity

Susskind opens with a deliberately familiar philosophical slogan: big things are made of little things, and little things are made of littler things. He then adds the more ambitious version of the same idea, the one that really matters for fundamental physics: as one descends through deeper layers, the objects should become not only smaller but also simpler. His metaphor is that houses are made of bricks, and nobody mistakes a house for a brick. The brick is smaller, but it is also simpler.

The first move of the lecture is to confront that expectation with the actual state of particle physics. If reductionism were delivering increasing simplicity in that strong sense, one might expect present-day elementary particle physics to be close to economical. Instead, Susskind emphasizes that even before quantum gravity enters, the bookkeeping already looks messy. In the lecture these are rough counts, but the roughness is part of the point:

$$N_{\text{species}} \sim 75, \qquad N_{\text{independent}} \sim 10\text{–}20, \qquad N_{\text{parameters}} \sim 20. \tag{1.1}$$

Even this is not the end of the story. The Standard Model does not contain gravity, does not explain dark matter, does not supply the extra structure needed for inflation, and carries a severe fine-tuning problem. Once one adds the "bells and whistles" that are usually invoked to repair these omissions, the effective count swells rather than shrinks. In the lecture supersymmetry is the clean example, and the schematic message is simply

$$N_{\text{parameters}}^{\text{extended}} \sim 200. \tag{1.2}$$

This quantitative complaint is not yet Susskind's main thesis. It is the prelude. The deeper claim, and the one that drives the rest of the lecture, is that modern quantum field theory and string

theory do not obviously support a unique, coupling-independent distinction between the bricks and the houses. The next examples are chosen to show that the very meaning of "elementary" may depend on which variables are efficient.

1.2 Bosonization: A First Breakdown of Fixed Fundamentality

Susskind's first theoretical example comes from $1+1$-dimensional quantum field theory: one dimension of space, one of time. He announces at once that he is not going to derive the equivalence in detail; the point is not the formal proof but the conceptual lesson.

One starts with a field theory of fermions. The fermion field is denoted by ψ. Two nearby fermions can form a bosonic composite, and the corresponding bosonic field is denoted by ϕ. At first sight the hierarchy seems straightforward: ψ is basic, while ϕ is a composite or effective variable. In the lecture the boson is introduced exactly this way, as a field that creates bound pairs of fermions.

What was discovered, however, is that the same theory can be rewritten from the opposite starting point. One may begin with the bosonic field ϕ, and then the fermion no longer appears as a point-like constituent. Instead it appears as a kink of the bosonic field, an extended configuration in which ϕ interpolates smoothly between two asymptotic values. The minimum mathematical content of the lecture is this asymptotic structure:

$$\phi(x \to -\infty) = \phi_-, \qquad \phi(x \to +\infty) = \phi_+. \tag{1.3}$$

A convenient schematic representative is

$$\phi(x) \approx \frac{\phi_+ + \phi_-}{2} + \frac{\phi_+ - \phi_-}{2} \tanh\left(\frac{x}{\ell}\right), \tag{1.4}$$

where ℓ is the width of the transition region. This is a standard reconstruction of the picture Susskind gives on the board; it is not meant to import a full sine-Gordon/Thirring derivation into a lecture that does not provide one.

The belt analogy in the lecture is precise enough to be worth keeping. A flat belt with no twist is a configuration with no kink. If one rotates one end by 2π, the belt returns to the same local orientation far away, but in between it must pass through an extended twisted region. That twisted region is the kink. In the bosonic description, the fermion is represented by precisely such an extended object.

The point becomes sharper once a coupling is introduced. Susskind calls it G in this example. The statement is not that the two descriptions are vaguely related, but that the efficiency of the two descriptions depends on the parameter regime:

$$G \ll 1 \Rightarrow \ell \text{ small, the kink is tight, and the } \psi\text{-description is efficient,} \tag{1.5}$$

$$G \gg 1 \Rightarrow \ell \text{ large, the kink spreads out, and the } \phi\text{-description is efficient.} \tag{1.6}$$

For small G, the kink is tight and the fermion looks like the simple object. As G is increased, the kink broadens, the fermion becomes less point-like, and the bosonic variables become the more economical ones.

Schematic update rule.

1. Begin with a fermionic theory described by ψ.
2. Identify bosonic composites described by ϕ.
3. Rewrite the same theory entirely in terms of ϕ.
4. In the bosonic description, the fermion reappears as a kink of width ℓ.
5. For $G \ll 1$, ℓ is small and the fermionic description is simplest.
6. For larger G, ℓ grows and the bosonic description becomes simplest.

This is the first explicit counterexample to the fixed "houses and bricks" picture. The lecture's punchline is not merely that composite objects exist, but that what counts as the elementary variable can change with the coupling.

1.3 Electric Charge, Magnetic Monopoles, and Weak versus Strong Coupling

Susskind next asks whether an analogous story appears in more familiar particle physics. The example is electric charge versus magnetic monopole. An ordinary charged particle, such as the electron, is an electric monopole in this language: a localized source of electric field. The magnetic monopole is introduced by the bar-magnet analogy. If one could somehow isolate the end of a bar magnet without the invisible connecting bar, the field lines would look exactly like those of an electric charge, but magnetic rather than electric.

The relevant control parameter is the fine-structure constant α. In the lecture Susskind treats it heuristically as the square of the electric charge:

$$\alpha \sim e^2. \tag{1.7}$$

He also gives a rough operational interpretation: it measures how readily an electron radiates, and he summarizes that strength by saying that an accelerated electron radiates on the order of "1/100" photons in the relevant sense. In the lecture's own rough numerical language,

$$\alpha \approx 10^{-2}. \tag{1.8}$$

This is not intended as a precision convention; it is a pedagogical way of encoding that electric charge is weak.

When $\alpha \ll 1$, the electron is the simple object. Its electric field is weak, its vacuum polarization cloud is dilute, and it looks almost point-like. The magnetic monopole is then the opposite: its field is enormous, its surroundings are violently distorted, and it is broad, complicated, and heavy. Susskind summarizes the situation in precisely this weak–strong style:

$$\alpha \ll 1 \Rightarrow \text{electron simple, monopole heavy and extended,} \tag{1.9}$$

$$\alpha \gg 1 \Rightarrow \text{monopole simple, electron heavy and structured.} \tag{1.10}$$

The lecture does not derive a monopole mass formula. It does not need one. Its point is that as one increases α, the electron's field becomes stronger, the electron becomes surrounded by more structure, and the monopole shrinks and lightens. If α were pushed far past unity, the convenient description would exchange: the monopole would be the simple object and the electron the complicated one.

That is the field-theory version of the earlier lesson from bosonization. The distinction between basic and composite is not absolute. It depends on the coupling.

A question about calculus. At this point a student asks whether calculus itself is an intrinsically reductionist tool. Susskind's answer is important because it briefly restates the whole philosophical point in different language. Calculus, he says, does not by itself tell us what the fundamental objects are; it only encodes smooth variation. The problem is that quantum fields do not behave like smooth classical fields on every scale. Rather, they fluctuate on all scales. There is no final clean layer at which the continuum simply reveals a unique set of smallest building blocks. This answer serves as a bridge: it closes the field-theory part of the argument and prepares the move to string theory.

1.4 Strings, D-Branes, and the Return of Weak–Strong Duality

The next transition in the lecture is deliberately naive. If string theory is supposed to be the theory of the fundamental building blocks, then surely the answer must be simple: the building blocks are strings. Susskind states that expectation openly, and then spends the next part of the lecture dismantling it.

1.4.1 Why strings are not the whole story

The first complication is the unavoidable presence of D-branes. A Dp-brane is a p-dimensional object on which strings can end. The letter D comes from Dirichlet boundary conditions, but the name is far less important than the role: these objects are part of the theory and cannot be discarded. They are not optional ornaments.

The lecture begins with the simplest examples:

- a D0-brane, which is point-like,
- a D1-brane, or D-string, which is string-like,
- an F-string, the ordinary fundamental string.

A D0-brane is a heavy point-like site on which any number of F-strings can end. A D1-brane is a heavy cable-like object, much thicker and more massive per unit length than an F-string, but likewise a place where F-strings can end. At weak coupling the lecture does not present these objects as equally elementary alternatives. The D-branes are introduced as heavy, structured, composite-looking objects that the theory nevertheless forces upon us.

Notation. Susskind uses G and g somewhat loosely in different parts of the lecture. In these notes G is reserved for the bosonization example, while g denotes the string coupling.

The string coupling g is defined operationally, not axiomatically. It is the parameter that measures the probability that a fluctuating string splits and forms daughter strings. This is why Susskind compares it directly to the role of α in QED. The lecture also pauses to identify tension with energy per unit length:

$$T = \frac{E}{L} = \frac{M}{L}. \tag{1.11}$$

1.4.2 F-strings and D-strings

The weak–strong story then repeats itself in string language. At small coupling the F-string is light, thin, and close to the original intuitive image of the fundamental object. The D-string is thick, heavy, and full of F-string structure. As the coupling grows, the F-string splits more readily; the cloud of daughter strings proliferates around it, and it becomes heavier and more complicated. At the same time the D-string has increasing difficulty emitting the now-heavy F-strings, so its surrounding structure thins out and it becomes lighter and more line-like. The lecture's weak–strong map is therefore

$$g \ll 1 \Rightarrow \text{F-string light and thin,} \quad \text{D-string heavy and structured,} \tag{1.12}$$

$$g \sim 1 \Rightarrow \text{F-string and D-string look comparable,} \tag{1.13}$$

$$g \gg 1 \Rightarrow \text{D-string light and thin,} \quad \text{F-string heavy and structured.} \tag{1.14}$$

Susskind remarks that around $g \sim 1$ the two descriptions cease to be sharply distinct, and beyond that point they effectively exchange roles. This is the string-theoretic version of the electron–monopole interchange. The accepted name for this is S-duality.

The lecture pushes the point one step further. In the idealized theory the coupling is not merely a number to be dialed by hand; it behaves like a field. Schematically,

$$g = g(x). \tag{1.15}$$

Different regions of space can therefore favor different dual descriptions. In one region F-strings are the light variables; in another region D-strings are. In between, nothing is especially simple. That is why, for Susskind, string theory does not restore the simple reductionist picture. It radicalizes the ambiguity.

1.4.3 A question-driven return to quantum electrodynamics

A student then asks whether the larger splitting probability simply means a more complicated structure "upstream." Susskind answers by returning to quantum electrodynamics and drawing the electron's worldline again. This return matters: it shows that the string story is not an isolated curiosity but another instance of the same weak–strong pattern.

The lecture's point is that even a point electron is accompanied by a hierarchy of processes:

1. an electron emits and absorbs photons,
2. a photon can create an electron–positron pair,
3. the daughter charged lines can themselves emit photons,
4. the hierarchy continues to smaller scales.

For weak coupling this cloud is dilute, so the electron looks almost point-like in the laboratory. For stronger coupling it proliferates, and the electron becomes a structured object. The string story is then presented as a direct analogue of this.

The same later discussion also motivates Susskind's heuristic energy bookkeeping. He insists that the relevant energy is not merely the sum of free-particle contributions. One must also include interaction energy:

$$E_{\text{tot}} = E_e + E_\gamma + E_{\text{int}}. \tag{1.16}$$

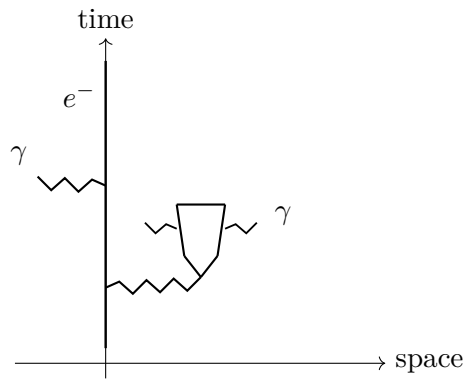


Figure 1.1: Transcript-based redraw of Susskind's later QED recap. A point electron emits photons, photons create pairs, and the daughter charged lines radiate again, producing an endless hierarchy of finer structure.

Here E_{int} is the crucial term. It is what lets the mass of the dressed electron include the energy associated with its surrounding field and virtual structure. The lecture does not turn this into a formal renormalized self-energy derivation, and neither should these notes. The point is heuristic but essential: the object's mass already includes the interaction energy required to support the complicated cloud around it.

The reason this figure belongs here, even though the weak–strong comparison was introduced earlier, is that in the lecture Susskind redraws the contrast at this later question-driven moment while discussing interaction energy and proliferating structure.

1.5 Even-Brane Theories and the Emergent Extra Dimension

Up to this point the lecture has been telling a strong story about exchange: fermion versus boson, electron versus monopole, F-string versus D-string. The next step is more surprising. Susskind switches to a different version of string theory, one with even-dimensional D-branes rather than odd-dimensional ones. The distinction matters because in this theory there is no D-string to exchange with the F-string. One has D0-branes, D2-branes, D4-branes, and so on.

The strong-coupling limit therefore cannot simply be a relabeling of one string as another. According to the lecture, something stranger happens: an extra spatial dimension becomes manifest. Susskind deliberately states this in the most dramatic form first — "a new dimension materializes" — and then immediately translates it into the more careful statement: a very small compact direction expands until it can no longer be ignored.

The bookkeeping is part of the point, so he pauses over it:

$$\text{10-dimensional string theory} = 9 \text{ space} + 1 \text{ time}, \qquad \text{11-dimensional theory} = 10 \text{ space} + 1 \text{ time}. \tag{1.17}$$

The lecture's claim is that the strong-coupling limit of the even-brane theory is effectively eleven-dimensional.

1.5.1 The peanut-butter sandwich

To picture the compact direction, Susskind uses the "peanut-butter sandwich" model: two infinitely thin pieces of bread, with real space between them. The special rule is that going out one side returns one to the other. In a conventional notation that the lecture describes pictorially rather than formally, one may write

$$y \sim y + 2\pi R. \tag{1.18}$$

If R is very small, observers living in the sandwich effectively describe their world as lower-dimensional. The key strong-coupling update is then

$$g \uparrow \qquad \Longrightarrow \qquad R_{\text{compact}} \uparrow . \tag{1.19}$$

That is what Susskind means when he says that a new dimension of space appears: not that space is created from nothing, but that a previously tiny compact direction expands and becomes visible in the effective theory.

1.5.2 What becomes of the D0- and D2-branes?

The lecture now asks what happens to the D0-branes and the strings. The answer is asymmetric. As g becomes large, the D0-branes get lighter and lighter; in the higher-dimensional interpretation they are simply gravitons carrying momentum in the newly manifest direction. Thus a heavy, complicated object in the weak-coupling picture becomes an ordinary light particle in the strong-coupling picture.

The D2-branes explain the fate of the strings. There are two configurations emphasized in the lecture:

1. a D2-brane floating between floor and ceiling like a carpet in the peanut butter;
2. a D2-brane stretched from floor to ceiling like a ribbon, extended along the visible direction.

When the sandwich is thin, the ribbon looks effectively one-dimensional. It is the object that low-energy observers call a string. As the compact direction opens up, that same object becomes unmistakably membrane-like. Because it now contains more material in the vertical direction, it becomes heavy and unstring-like.

The logic of the transition.

1. Start with a very thin compact direction, so the floor-to-ceiling ribbon behaves effectively like a string.
2. Increase g, and the compact direction expands.
3. The D0-branes become light and are reinterpreted as gravitons in the higher-dimensional theory.
4. The string-like D2 ribbon becomes a visibly extended membrane and grows heavy.

This is the lecture's hinge from duality to emergent geometry. Strong coupling does not merely exchange one object for another; it changes which geometry is the efficient one.

1.6 D3-Branes, Wrapped Membranes, and a Geometrical Picture of Duality

After the break Susskind explicitly describes the subject as an "enormous web" of interconnected theories and ideas. The next examples are not new isolated effects; they are bridges tying together the previous ones.

1.6.1 A D3-brane worldvolume

The first bridge uses a D3-brane. A D3-brane has three spatial dimensions, and an observer confined to it can easily mistake it for ordinary space. Strings can end on it, and from the worldvolume point of view the endpoint of a string looks like a particle.

This is one of the lecture's most satisfying identifications. An F-string ending on the D3-brane appears to the worldvolume observer as an electric charge. A D-string ending on the same brane appears as a magnetic monopole. In other words, the duality between F-strings and D-strings becomes, on the D3-brane worldvolume, the electric–magnetic duality of ordinary field theory. What was earlier an analogy is now an explicit dictionary.

1.6.2 Two compact directions and asymmetric wrapping

Susskind then returns to the higher-dimensional geometry and compactifies not one but two small directions. Since one cannot draw eleven dimensions on a blackboard, he compresses the picture to one visible large direction and two compact ones. He insists on drawing the compactification asymmetrically, because one direction must be shorter than the other if the effective strings are to have different tensions.

The clean mathematical content is then as follows. A membrane may wrap one compact direction and produce one effective string, or it may wrap the other compact direction and produce another. Since the compact directions do not have the same size, the two effective strings do not have the same tension. If R_1 and R_2 denote the two compact scales, then the lecture's later interpretation is that the weak–strong parameter is encoded geometrically as the aspect ratio

$$\alpha \sim \frac{R_1}{R_2}. \tag{1.20}$$

Here the notation shifts: α is no longer the QED fine-structure constant. It now denotes a geometric ratio. The lecture says this explicitly in response to a student question, and the shift should be flagged because the same symbol is being reused for a different purpose.

Remark 1.1. In the geometric part of the lecture, α is being reinterpreted as an aspect-ratio parameter. The symbol is the same as before, but the meaning is different.

A clean transcript-backed reconstruction of the wrapped-membrane picture is:

This is the geometric version of the earlier duality story. Squeezing one direction while stretching the other makes one wrapped object heavier and the other lighter, so the roles of F-string and D-string can interchange.

At this point a student asks why the picture is not a helix rather than a circle. Susskind's answer is revealing. The geometry should be thought of as a compact direction cut open and drawn flat, so

that exiting one side means re-entering at the identified side. The lecture emphasizes that this is a geometrical picture, useful but not exhaustive. It organizes the duality, but it does not by itself explain every microscopic feature of how a D-brane, under sufficiently hard probing, can look like a collection of F-strings.

1.7 Idealization, Specially Symmetric Points, and the Landscape

The last movement of the lecture steps back from the elegant duality examples and asks what they do, and do not, tell us about the real world.

First, Susskind warns that the precisely controlled dualities are features of highly idealized supersymmetric theories. His analogy is to circular orbits in Newtonian mechanics: one studies them first because they are symmetric and mathematically tractable, not because actual planetary motion is exhausted by perfect circles. In exactly the same way, supersymmetric string theories are easier to analyze because they are more symmetric than nature appears to be.

Within those idealized theories there are specially symmetric points. In the geometric picture these are the equal-aspect configurations,

$$R_1 = R_2, \tag{1.21}$$

at which the two wrapped descriptions coincide. The corresponding field-theory statement is that the electric and magnetic descriptions are equally simple:

$$m_{\text{monopole}} = m_e. \tag{1.22}$$

Susskind immediately adds the physical caveat that our world does not seem to live at such a point; it is lopsided rather than symmetric.

This leads naturally into the lecture's final large-scale moral. The examples one can describe cleanly are special precisely because they are special. The generic construction in a vast configuration space need not be elegant. Susskind illustrates this first with coding theory and then with DNA: mathematically structured examples are easy to analyze, but the space of possibilities is overwhelmingly larger than the space of neat examples. The same combinatorial point then becomes the "landscape" argument of string theory.

Suppose a compact space has roughly 500 cycles, and each cycle admits on the order of 10 different choices of wrapped branes, fluxes, or related structures. Then the number of resulting constructions grows schematically like

$$N_{\text{vacua}} \sim 10^{500}. \tag{1.23}$$

The lecture's counting logic is elementary:

$$10 \times 10 \times \cdots \times 10 = 10^{500} \tag{1.24}$$

with 500 factors. The DNA analogy makes the same point in a more familiar register:

$$N_{\text{DNA}}(500) = 4^{500}. \tag{1.25}$$

The exact exponents are not the point. The point is that simple local rules can generate a staggeringly large space of globally different constructions.

This gives Susskind's final inversion of the opening reductionist expectation. The complication of particle physics may not mean that the microscopic rules themselves are elaborate. It may instead

mean that the vacuum we inhabit is one complicated member of an enormous family built from relatively simple ingredients.

The last coda of the lecture turns briefly toward cosmology. A compact spatial direction that is sufficiently large is observationally hard to distinguish from an infinite one. Flatness, by itself, only gives lower bounds on size; it does not prove non-compactness. At the same time, Susskind stresses that the smoothly varying couplings of the idealized examples are not compatible with our world. If couplings vary at all in nature, the variations must be step-like rather than smooth, a point he postpones to later cosmological discussions.

1.8 Summary

The lecture begins with the philosophical hope that deeper layers of physics should be simpler and ends by showing how fragile that hope has become. In $1+1$-dimensional bosonization, the same theory can be organized either around fermions or around bosons, with the fermion reappearing as a kink. In quantum electrodynamics, weak and strong coupling exchange the descriptive roles of electron and magnetic monopole. In string theory, F-strings and D-strings repeat the same pattern, and in the even-brane theory strong coupling goes further still: a compact direction expands, D0-branes become gravitons, and the effective strings are reinterpreted as wrapped membranes in a higher-dimensional geometry. The final lesson is that modern quantum field theory and string theory do not furnish a unique once-and-for-all answer to the question "which are the bricks and which are the houses?" They furnish a web of descriptions, each efficient in its own regime, together with a huge landscape of possible constructions whose generic output is complexity rather than simplicity.

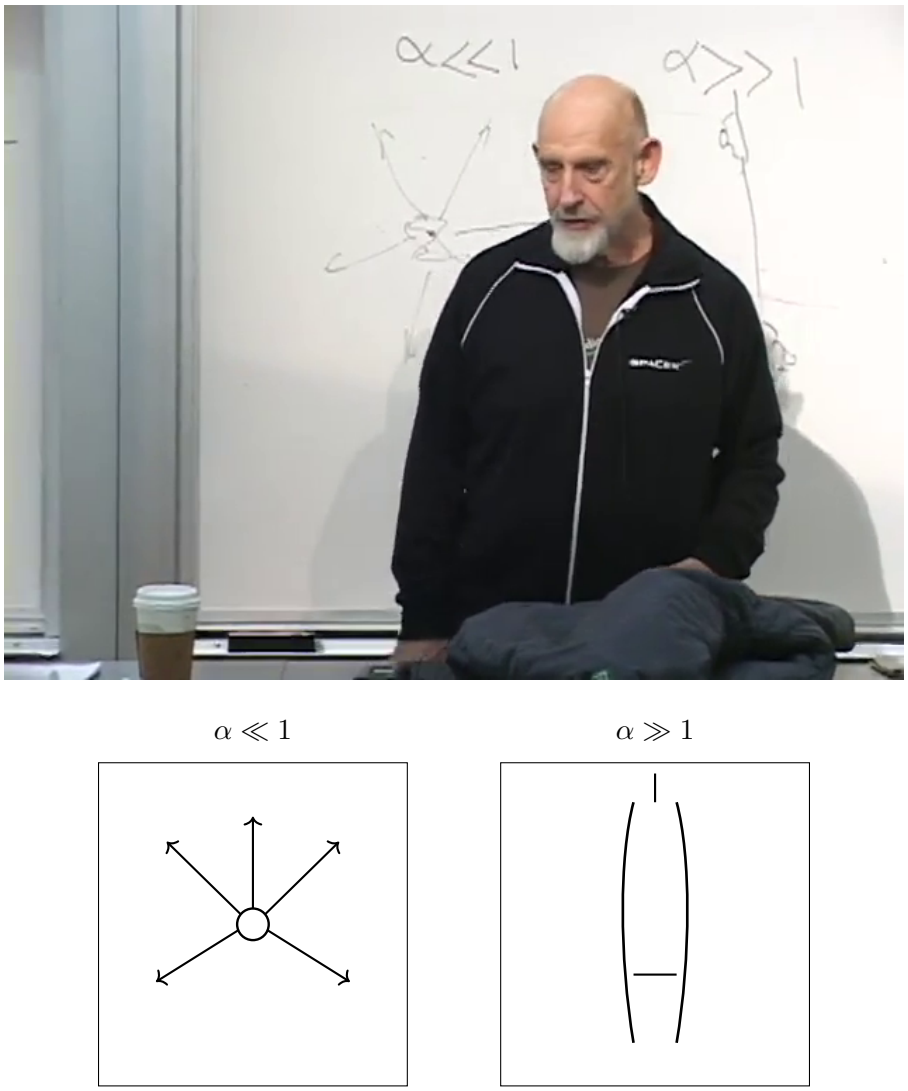


Figure 1.2: A later board recap of the weak–strong QED comparison. The screenshot is kept as evidence for the blackboard layout; the redraw records only the robust contrast between a compact weak-coupling object and an extended strong-coupling one.

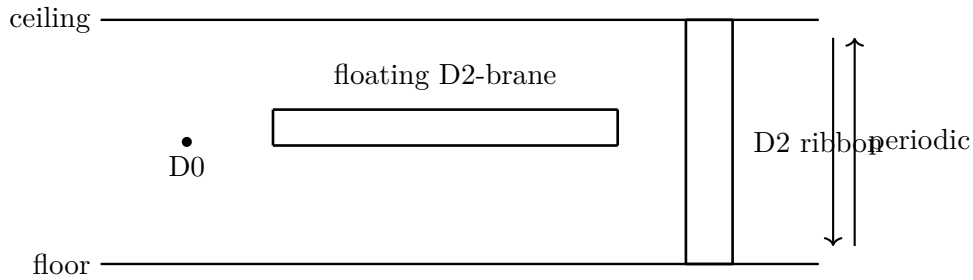


Figure 1.3: Transcript-based redraw of the sandwich compactification. The upper and lower boundaries are periodically identified; one D2-brane floats between them like a carpet, while another stretches from floor to ceiling and looks string-like when the compact direction is small.

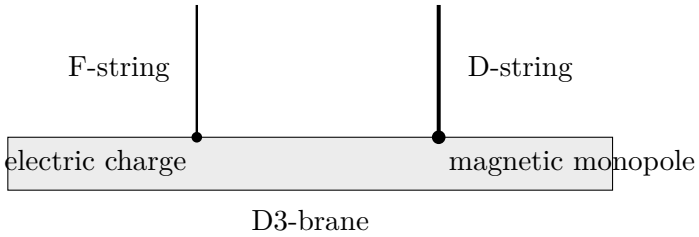


Figure 1.4: Transcript-based redraw of the D3-brane worldvolume interpretation. An F-string endpoint appears as an electrically charged particle; a D-string endpoint appears as a magnetic monopole.

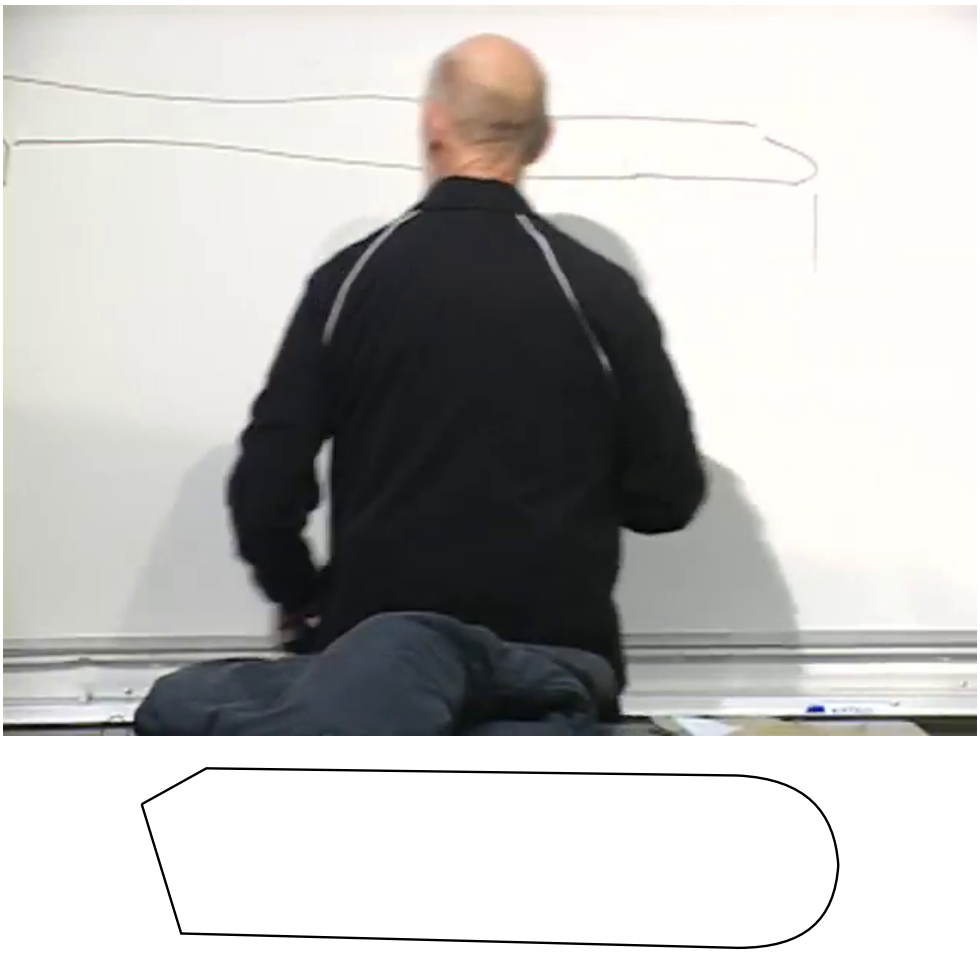


Figure 1.5: The asymmetric profile introduced when Susskind says he will draw the picture asymmetrically. The screenshot is kept because the argument here is strongly visual; the redraw regularizes only the large-scale asymmetry.

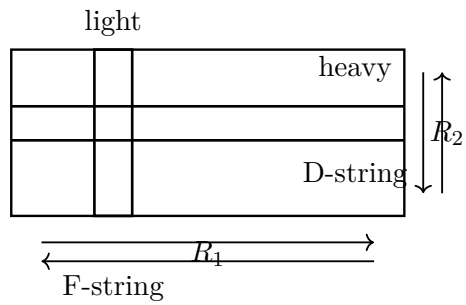


Figure 1.6: Clean reconstruction of the two-cycle picture. Two wrapped membrane configurations both look string-like from the non-compact directions, but unequal compact scales make one effective string lighter than the other.

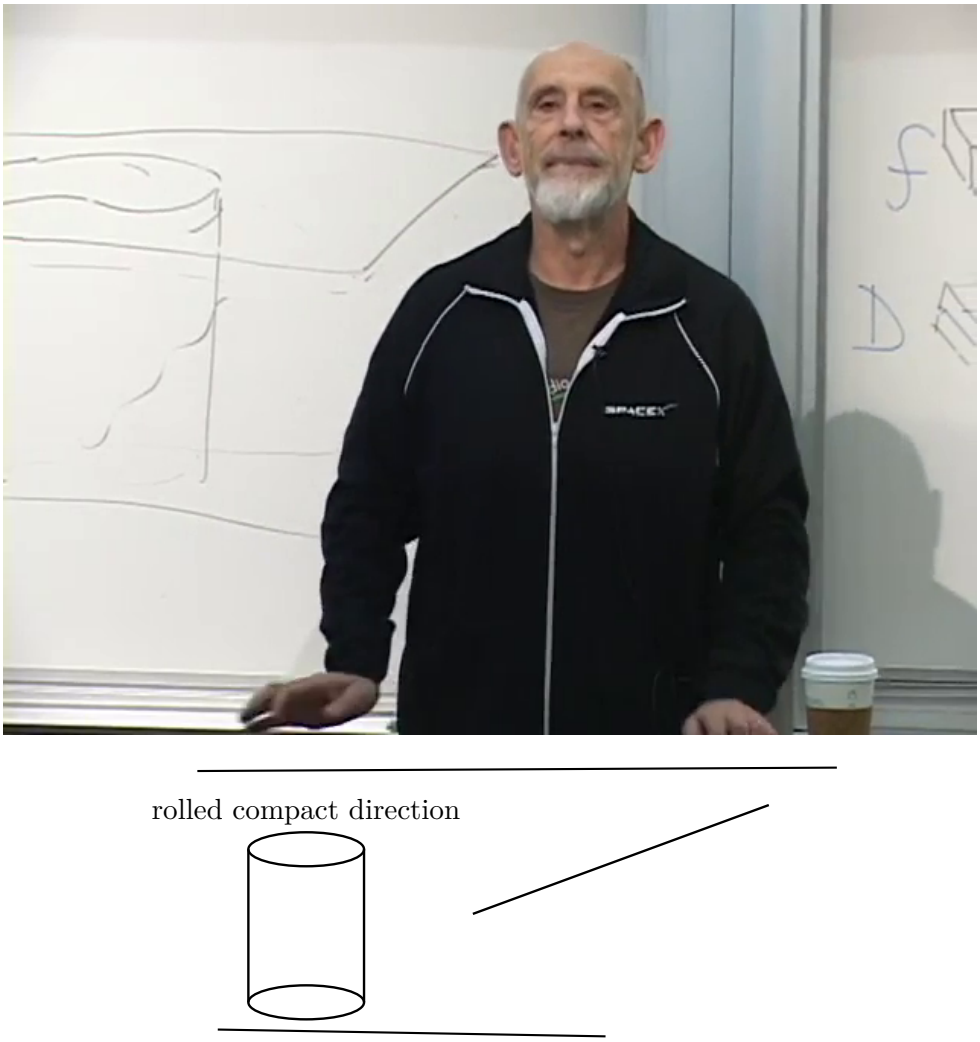


Figure 1.7: The lecture's "geometrical picture." The screenshot is kept as evidence for the actual board state; the redraw records only the cylinder-like compact direction, the long upper contour, and the slanted continuation that survive the occlusion.

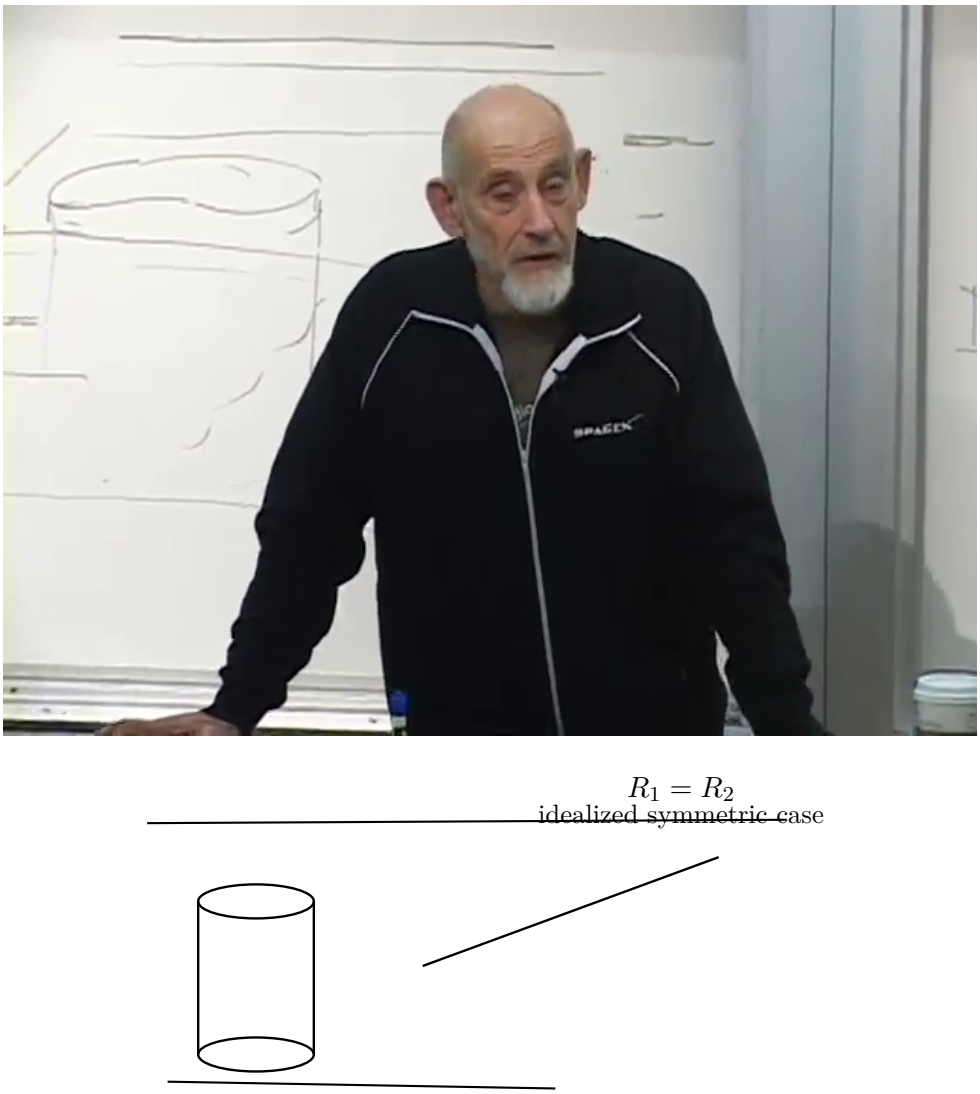


Figure 1.8: Later board frame retained during the discussion of specially symmetric points. The equal-aspect annotation is transcript-backed interpretation, not a direct transcription of visible chalk.

Chapter 2

Lecture 2: Topics in String Theory

Leonard Susskind opens this lecture by stepping back from cosmology and returning, for a moment, to the special theory of relativity. The point of the reset is not historical; it is methodological. Before black-hole geometry appears, he wants one simple operational rule in hand: light rays are identified by the condition of zero proper time. The lecture then carries that rule into curved spacetime, writes down the Schwarzschild metric, separates the horizon from the singularity, and only after that classical groundwork pivots toward the quantum mechanics of black holes, hidden information, and the Bekenstein-Hawking area law. These notes follow Leonard Susskind's presentation closely, with conservative normalization of notation where needed, and are curated by LazyingArt LLC.

2.1 Flat Spacetime, Proper Time, and Null Trajectories

Susskind begins with the metric of flat spacetime. For two neighboring events separated by dt, dx, dy, dz, the proper time is defined by

$$d\tau^2 = dt^2 - \frac{1}{c^2}\left(dx^2 + dy^2 + dz^2\right). \tag{2.1}$$

This is the relativistic interval. The spatial terms come with the opposite sign from the time term, so the geometry is not the Euclidean geometry of an ordinary four-dimensional space.

The factor $1/c^2$ is only a matter of units. One may always measure space and time in compatible units and set $c = 1$. Susskind keeps c for a moment in order to make the relation to the speed of light explicit, and then later suppresses it.

He also remarks that $d\tau^2$ need not be positive for arbitrary pairs of neighboring events. If the spatial separation is sufficiently large compared with the time separation, the expression becomes negative. He does not pursue that point here. Instead, he turns immediately to the question that matters for the rest of the lecture: how do light rays move?

For a light ray moving purely along the x-direction, one may set $dy = dz = 0$. The defining rule is then

$$d\tau^2 = 0. \tag{2.2}$$

This is the null condition. It is the lecture's first main principle, and it will survive every later change of geometry.

Applying it in flat spacetime gives

$$0 = dt^2 - \frac{1}{c^2}dx^2, \tag{2.3}$$

$$c^2dt^2 = dx^2, \tag{2.4}$$

$$c\,dt = \pm dx. \tag{2.5}$$

Dividing by dt yields

$$\frac{dx}{dt} = \pm c. \tag{2.6}$$

The plus and minus signs correspond to right-moving and left-moving light rays.

What matters most is not the particular number c, but the invariant statement behind it. In any metric, the way to determine the motion of light is to impose the condition that the interval along the trajectory vanish.

2.2 From the Metric Tensor to Spherical Coordinates

The lecture next broadens from special relativity to general relativity. In a general spacetime the geometry varies from place to place, and Susskind describes it by a metric tensor $G_{\mu\nu}$. We will use the standard notation

$$ds^2 = g_{\mu\nu}\,dx^\mu dx^\nu. \tag{2.7}$$

As usual, repeated indices are summed. This compact expression stands for all quadratic combinations of the coordinate differentials with their appropriate metric coefficients.

The logical rule for light does not change. In flat or curved spacetime alike, light rays are diagnosed by setting the interval to zero. Susskind alternates somewhat loosely between ds^2, $d\tau^2$, and the phrase "proper time," but the intended content is the same: light moves on null trajectories.

Because the black hole to be studied is spherically symmetric, he then rewrites flat space in spherical coordinates. Instead of Cartesian coordinates x, y, z, one uses a radial distance r and two angles. The angular geometry is summarized by the metric on the unit two-sphere,

$$d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2. \tag{2.8}$$

If the sphere has radius r, the corresponding angular line element is $r^2d\Omega^2$. Thus, in units with $c = 1$, flat spacetime may be written as

$$d\tau^2 = dt^2 - \left(dr^2 + r^2d\Omega^2\right). \tag{2.9}$$

This is the baseline against which the black-hole metric will be presented.

2.3 The Schwarzschild Metric as a Deformation of Flat Space

At this point Susskind announces that he will not solve Einstein's equations. He simply writes down the metric of a black hole and inspects it. The lecture's rhythm matters here: the metric is introduced not as an isolated formula but as a deformation of the flat spherical line element just discussed.

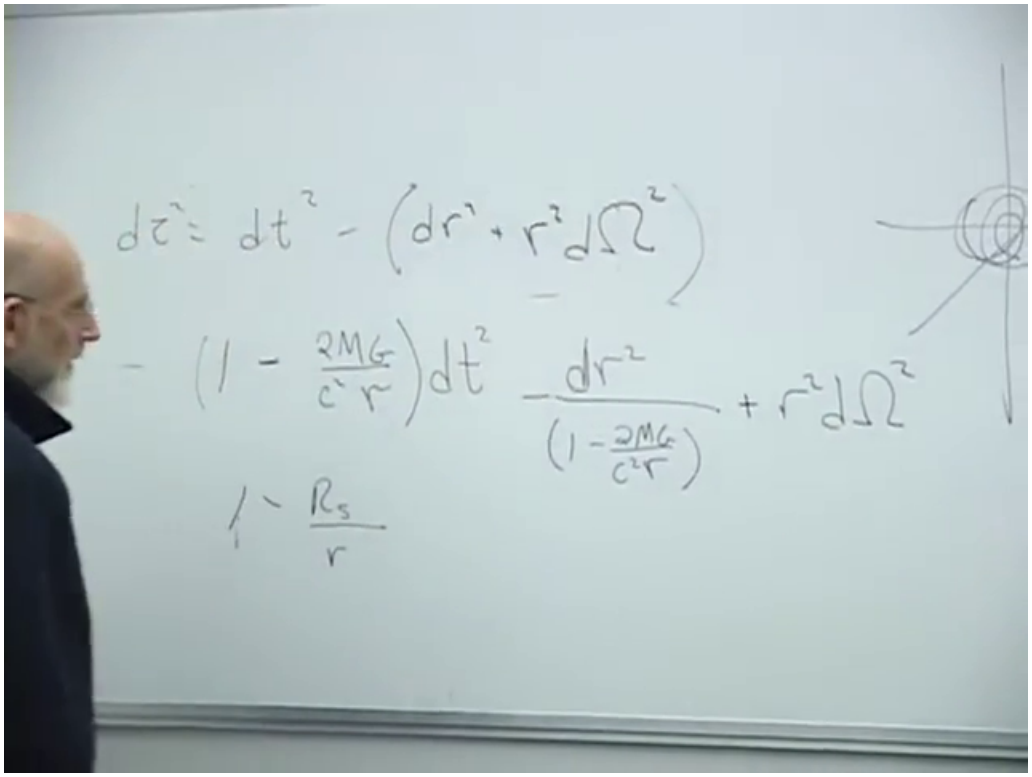


Figure 2.1: Schwarzschild metric and radial black-hole sketch. The board places the flat spherical line element above the Schwarzschild deformation and isolates the factor $1 - \frac{2MG}{c^2 r}$ together with the shorthand $1 - \frac{R_s}{r}$.

The board at this moment is worth preserving because it shows that comparison explicitly.

The upper line on the board repeats flat spacetime in spherical coordinates,

$$d\tau^2 = dt^2 - \left(dr^2 + r^2 d\Omega^2\right). \tag{2.10}$$

The lower line introduces the Schwarzschild factor. If the mass of the black hole is m, then its Schwarzschild radius is

$$R_s = \frac{2Gm}{c^2}, \tag{2.11}$$

so that the controlling factor may be written as

$$1 - \frac{2Gm}{c^2 r} = 1 - \frac{R_s}{r}. \tag{2.12}$$

Following the lecture, we keep c explicit once and then mostly set $c = 1$. In standard form, the Schwarzschild metric is

$$d\tau^2 = \left(1 - \frac{2Gm}{c^2 r}\right) dt^2 - \frac{dr^2}{1 - \frac{2Gm}{c^2 r}} - r^2 d\Omega^2. \tag{2.13}$$

Remark 2.1. The screenshot strongly supports the structure of this equation, especially the repeated factor $1 - \frac{2MG}{c^2 r}$ and the shorthand $1 - \frac{R_s}{r}$. However, the handwritten lower line is partly cropped and not every sign is perfectly legible. The formula above is therefore a cautious standard reconstruction, chosen because it matches the transcript and reduces to the flat spherical metric for $r \to \infty$.

That asymptotic check is the first thing Susskind emphasizes. Far from the black hole, $R_s/r \ll 1$, so the coefficient of dt^2 tends to 1, the denominator in the radial term tends to 1, and the metric becomes the flat spherical line element again.

The right-hand side of the board also carries a small unlabeled radial sketch. It is schematic rather than analytic, but it is useful evidence for the way spherical symmetry is being visualized.

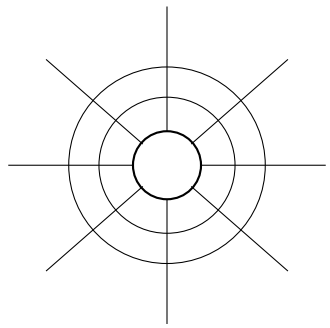


Figure 2.2: A minimal redraw of the board's radial sketch. It should be read only as a schematic cue for spherical symmetry.

2.4 Horizon, Singularity, and the Motion of Radial Light

Once the metric is on the board, the lecture immediately asks where it looks strange. From here on Susskind suppresses c most of the time, so $R_s = 2Gm$. Two radii stand out:

$$r = 2Gm \qquad \text{and} \qquad r = 0. \tag{2.14}$$

He is careful not to identify them with one another. The radius $r = 2Gm$ is the horizon. The radius $r = 0$ is the singularity. These are physically very different.

The difference is explained through tidal forces. At the horizon the tidal forces are finite, and for a sufficiently large black hole they are small. Falling through the horizon need not be violent at all. At the singularity, by contrast, tidal forces become infinite. That is the genuinely destructive place.

Only after making that distinction does Susskind ask again how light moves. Consider a radial light ray. Along such a ray the direction on the sphere is fixed, so

$$d\Omega^2 = 0. \tag{2.15}$$

The rule is still the same as in flat spacetime: along a light ray, the proper time is zero. Applying that rule to the Schwarzschild metric gives

$$0 = \left(1 - \frac{2Gm}{r}\right) dt^2 - \frac{dr^2}{1 - \frac{2Gm}{r}}. \tag{2.16}$$

Rearranging,

$$\left(1 - \frac{2Gm}{r}\right) dt^2 = \frac{dr^2}{1 - \frac{2Gm}{r}}. \tag{2.17}$$

Multiplying through by the denominator and taking the square root yields

$$dr = \pm \left(1 - \frac{2Gm}{r}\right) dt, \tag{2.18}$$

and therefore

$$\frac{dr}{dt} = \pm\left(1 - \frac{2Gm}{r}\right). \tag{2.19}$$

The plus sign is the outgoing branch; the minus sign is the ingoing branch.

Remark 2.2. There is a short garbled stretch in the transcript during this manipulation, but the surrounding clean lines determine the algebra unambiguously. The reconstruction above is exactly the step needed to recover the light-ray equation that Susskind then discusses verbally.

This is the one explicit property of the metric that the lecture promised to extract. Far from the black hole, $2Gm/r \ll 1$, and the outgoing branch reduces to

$$\frac{dr}{dt} \approx 1, \tag{2.20}$$

which is just the $c = 1$ version of ordinary light motion. Near the horizon,

$$\frac{dr}{dt} \to 0 \qquad \text{as} \qquad r \to 2Gm. \tag{2.21}$$

In these coordinates, outward-moving light rays slow down and finally stand still at the horizon.

Susskind then translates this into a spacetime picture in the t-r plane: time vertical, radius horizontal, and a vertical line marking $r = 2Gm$. Outgoing light rays start near 45° far away but become more and more vertical as they approach the horizon. Ingoing rays, drawn in the same coordinates, asymptotically approach the horizon.

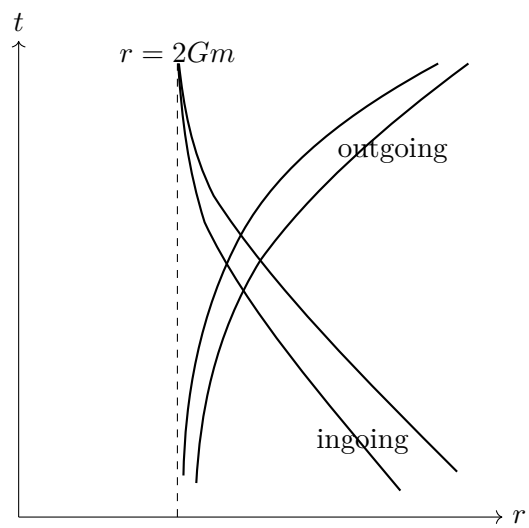


Figure 2.3: Coordinate-time picture of radial light rays reconstructed from the lecture. Outgoing rays flatten toward the horizon; ingoing rays asymptotically approach it in coordinate time.

A student immediately raises the obvious objection: does it really make sense to say that light slows down? Susskind's reply is important. The statement that light travels at "the speed of light" depends on coordinates and units. The invariant statement is that light follows null trajectories. What vanishes at the horizon is the coordinate velocity dr/dt in Schwarzschild coordinates, not the fundamental null character of the light ray.

The lecture closes the geometric part with one more strong claim. Inside the horizon, all light rays fall inward. Whether one tries to aim them outward or inward, they are carried toward smaller r, eventually toward $r = 0$.

2.5 From the Horizon to the Information Problem

With that coordinate picture in place, Susskind changes direction abruptly. He says that he wants to jump from the geometry of black holes to their quantum mechanics. The motivation for the jump comes directly from the horizon picture just developed.

From the outside point of view, matter falling toward a black hole becomes unavailable. One may imagine it as falling in, or one may imagine it as getting squeezed closer and closer to the horizon and effectively sticking there. Either way, the information it carries is no longer accessible to an outside observer.

By itself, that is not yet a contradiction. One might simply say that the information has not been destroyed; it has only become concentrated near the horizon. But the lecture immediately adds the crucial quantum claim: black holes are not perfectly dead classical objects. They have temperature, they radiate, and therefore they lose energy.

If a black hole radiates, then its mass decreases:

$$E = mc^2. \tag{2.22}$$

Since its horizon radius is

$$R_s = \frac{2Gm}{c^2}, \tag{2.23}$$

a decrease in mass means a decrease in horizon size. Given enough time, the black hole evaporates. That is the real puzzle. What happens to the information that had become trapped, or effectively trapped, at the horizon? Does it disappear? Does it remain in a remnant? Or does it somehow emerge again in the outgoing radiation?

Before answering those questions, Susskind backs up once more. If one wants to understand why black holes evaporate and why they have temperature, one first has to understand why they should be assigned an entropy at all.

2.6 Entropy as Hidden Information

The lecture now rebuilds entropy from the language of information. A system contains information insofar as one can distinguish it from other systems. Operationally, that means asking yes-no questions. The number of such yes-no questions needed to specify a state is its information content, measured in bits.

Susskind illustrates this with a box partitioned into smaller boxes. If a box is either empty or occupied, then a long string of yes-no answers describes the configuration. More generally, one may imagine an arbitrarily fine partition of space and enough questions to identify the particles present. In that language, information is the number of binary answers needed to distinguish one state from another.

A second principle follows immediately: information does not disappear. Two distinct states remain distinct under time evolution. What can happen is that information becomes hidden. This is the role of the bathtub example. Water dripped into a tub may initially carry visible patterns or ripples, but after enough time the macroscopic state looks featureless. The information has not been destroyed; it has been pushed into microscopic degrees of freedom that are too numerous and too small to keep track of.

Definition 2.3. In the lecture's language, entropy is hidden or inaccessible information: the number of bits that are present in a physical state but unavailable to the observer.

This point of view is then tied directly to thermodynamics. Susskind introduces the standard symbol S for entropy and interprets S in bit units. In that normalization, an increase of one unit of S means one additional hidden bit.

He next redefines temperature through the cost of hiding information. The example is Landauer's rule for a computer. Erasing a bit from a computer does not make information vanish; it transfers that bit into the environment. The environment heats up by a small amount. In that sense, temperature is the energy cost of hiding one more bit in a heat bath.

In the lecture's bit normalization, if $\Delta S = 1$, then the energy cost is

$$\Delta E = T. \tag{2.24}$$

For several bits one has $\Delta E = T\,\Delta S$, and in differential form this becomes

$$dE = T\,dS. \tag{2.25}$$

Susskind emphasizes that this is one of the basic formulas of thermodynamics, but that it becomes conceptually transparent only when entropy is understood first as hidden information. No Boltzmann constant is introduced here; the lecture stays in the bit-count normalization.

2.7 Bekenstein's One-Bit Argument and the Area Law

The final part of the lecture returns to the black hole itself. Instead of declaring that the black hole has entropy, Susskind follows Bekenstein's route: compute how much information can be hidden there. The organizing image is deliberately elementary. One can ask how full a bathtub becomes by adding drops one at a time; likewise one can ask how large a black hole becomes by adding bits one at a time.

That immediately raises a subtlety. Throwing in a generic object does not mean throwing in just one bit. A short-wavelength photon, for example, carries many distinguishable pieces of information because one may ask where on the horizon it entered. The arrival location alone can encode many yes-no answers.

To get as close as possible to one bit, one wants a photon whose position is maximally delocalized. In quantum mechanics, the relevant scale is its wavelength. If the wavelength is of order the size of the black hole, then the photon is so spread out that it carries essentially no positional information beyond the yes-no question of whether it entered. But the wavelength cannot be arbitrarily large: if it is much larger than the black hole, the photon will not enter effectively at all. Thus the lecture chooses

$$\lambda \sim R_s \sim 2Gm, \tag{2.26}$$

with the final step read in the $c = 1$ convention used through the geometric part of the lecture.

For such a photon, the energy is estimated by

$$E = cp, \qquad p \sim \frac{h}{\lambda}, \qquad E \sim \frac{ch}{\lambda}. \tag{2.27}$$

Susskind is explicit that this is only an order-of-magnitude estimate. He ignores factors of 2, 2π, and speaks somewhat loosely about h versus $\hbar$. The argument is meant to determine scaling, not the exact coefficient.

Substituting $\lambda \sim R_s$ gives the energy required to hide one extra bit:

$$\Delta E \sim \frac{ch}{R_s}. \tag{2.28}$$

At this point Susskind remarks that one could almost stop. In the lecture's normalization, that energy cost per bit is the temperature. Thus the black hole's temperature is of order

$$T_H \sim \frac{ch}{Gm}, \tag{2.29}$$

up to the numerical factors that this simple argument does not control.

He then continues, because the same estimate also tells us how the horizon grows. Since $E = mc^2$,

$$\Delta m \sim \frac{\Delta E}{c^2} \sim \frac{h}{R_s c}. \tag{2.30}$$

The Schwarzschild radius is proportional to Gm/c^2, so the corresponding change in radius is

$$\Delta R_s \sim \frac{G\,\Delta m}{c^2} \sim \frac{Gh}{R_s c^3}. \tag{2.31}$$

Now multiply by R_s. Since the horizon area scales like R_s^2, one finds

$$\Delta A \sim R_s\,\Delta R_s \sim \frac{Gh}{c^3}. \tag{2.32}$$

This is the crucial universal step in the argument: each additional bit changes the horizon area by the same tiny amount, independent of the size of the black hole at that moment.

That universal area scale is the Planck area,

$$\ell_P^2 \sim \frac{G\hbar}{c^3}. \tag{2.33}$$

The lecture's picture is then vivid: bits pile up at the horizon, each occupying about one Planck area. In that sense, the hidden information of a black hole is proportional not to its volume but to its area.

Thus Bekenstein's conclusion is

$$S_{\mathrm{BH}} \propto \frac{A\,c^3}{G\hbar}. \tag{2.34}$$

Susskind then adds the historical refinement that closes the lecture. Bekenstein identified the proportionality, but Hawking fixed the coefficient. The exact entropy is

$$S_{\mathrm{BH}} = \frac{A\,c^3}{4G\hbar}. \tag{2.35}$$

Two final morals are emphasized. First, the famous factor 1/4 is not obtained from the crude one-bit estimate; it is Hawking's refinement. Second, $\hbar$ sits in the denominator. That means quantum mechanics does not create black-hole entropy as a small quantum correction. Rather, it makes the entropy finite. In the classical limit $\hbar \to 0$, one could hide arbitrarily much information at arbitrarily small energy cost, and the entropy would diverge.

2.8 Summary

The lecture unfolds in a deliberately staged way. It begins with flat spacetime and the meaning of a metric, extracts from special relativity the null condition $d\tau^2 = 0$, and then carries that same rule into the Schwarzschild geometry. With that tool, Susskind distinguishes horizon from singularity and derives the coordinate-time behavior of radial light rays. Only then does he turn to the quantum problem: a black hole that can radiate must also carry entropy, and the natural language for that entropy is hidden information.

Bekenstein's one-bit argument provides the bridge. A photon of wavelength comparable to the horizon size adds roughly one hidden bit, changes the black-hole energy by an amount of order the Hawking temperature, and increases the horizon area by about one Planck area. The result is that black-hole entropy is proportional to area, with the exact formula $S_{\mathrm{BH}} = \frac{A\,c^3}{4G\hbar}$. The chapter therefore ends where the lecture ends: black holes are not merely geometric curiosities, but thermodynamic systems whose horizon stores a finite amount of information.

Chapter 3

Black Hole Horizons, Acceleration Horizons, and Evaporation

These notes follow Leonard Susskind's third lecture in the supplementary 2011 "Winter Topics in String Theory" track, curated by LazyingArt LLC. The lecture unfolds in two distinct movements. First, Susskind wants to make good on a physical claim from the previous meeting: the horizon of a sufficiently large black hole should feel locally like empty flat space, even though the Schwarzschild metric seems to misbehave at $r = 2MG$. Second, once that local geometry is under control, he turns back to black-hole thermodynamics and asks what temperature, negative specific heat, and evaporation are trying to tell us about the underlying microscopic theory.

3.1 Why the Horizon Should Be Boring

Susskind opens by stating the conclusion he wants the mathematics to explain. If a black hole is very large, then an observer who falls through the horizon should find nothing dramatic there. Locally, the geometry should look exactly, or almost exactly, like empty Minkowski space. The horizon is therefore supposed to be physically boring.

What makes this claim nontrivial is the familiar Schwarzschild metric. In Schwarzschild coordinates, one coefficient vanishes at $r = 2MG$ and another diverges there. The lecture begins by sharpening that tension rather than hiding it. The question is not whether the metric looks singular in these coordinates; it plainly does. The question is whether that behavior signals real physics or only a bad coordinate choice.

3.2 Metric Convention and the Schwarzschild Starting Point

Susskind next resets notation. Earlier lectures used $d\tau^2$, positive on timelike intervals. Here he flips the overall sign and works with ds^2, which he describes as a "proper distance squared" convention. Nothing essential changes, but the formulas now appear in the form he prefers to manipulate on the blackboard. Throughout these notes I write M and G, although the spoken lecture often says m and g.

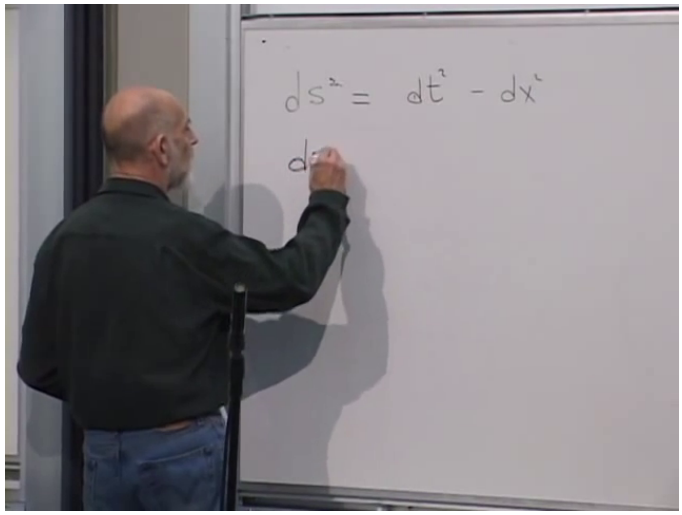


Figure 3.1: Board evidence for the sign-convention shift and the beginning of the Schwarzschild rewrite. The top line is a transitional reference line; the transcript controls the polished metric written below.

Remark 3.1. The visible board line

$$ds^2 = dt^2 - dx^2$$

should be read as a transitional flat-space reference during the sign change, not as the final governing convention for the lecture's black-hole formulas.

Suppressing the angular part exactly as Susskind does, the Schwarzschild metric is written as

$$ds^2 = -\left(1 - \frac{2MG}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2MG}{r}} + \cdots . \tag{3.1}$$

The omitted term is the angular piece $r^2 d\Omega_2^2$, which plays no role in the near-horizon derivation. Far from the hole, where $r \gg 2MG$, one has

$$ds^2 \approx -dt^2 + dr^2 + \cdots , \tag{3.2}$$

so the radial-time geometry is just the flat metric.

The trouble begins near $r = 2MG$. There the coefficient of dt^2 goes to zero, while the coefficient of dr^2 becomes large. Susskind's point is that this is only the beginning of the story. Before touching the black hole directly, he moves "over here" on the board and reviews a pair of coordinate systems whose whole purpose is to teach the audience not to overreact to singular-looking metric coefficients.

3.3 A Coordinate Warm-Up: Polar and Hyperbolic Polar Coordinates

Polar coordinates on the plane. On the ordinary Euclidean plane the metric is

$$ds^2 = dx^2 + dy^2. \tag{3.3}$$

Now introduce polar coordinates,

$$x = \rho\cos\theta, \qquad y = \rho\sin\theta. \tag{3.4}$$

A small displacement has one component $d\rho$ in the radial direction and one component $\rho\, d\theta$ in the transverse direction, so the metric becomes

$$ds^2 = d\rho^2 + \rho^2 d\theta^2. \tag{3.5}$$

At $\rho = 0$, the coefficient of $d\theta^2$ vanishes. But nothing singular happens at the origin. It merely means that changing θ at the origin does not move you anywhere. This is the first model of the phenomenon Susskind wants the class to keep in mind.

Hyperbolic polar coordinates in Minkowski space. He then replaces the Euclidean plane by $1+1$-dimensional Minkowski space,

$$ds^2 = -dt^2 + dx^2. \tag{3.6}$$

Again he introduces a radial variable ρ, but the angular variable is now a hyperbolic angle ω:

$$x = \rho\cosh\omega, \qquad t = \rho\sinh\omega. \tag{3.7}$$

The corresponding identity is no longer $\cos^2\theta + \sin^2\theta = 1$, but

$$\cosh^2\omega - \sinh^2\omega = 1. \tag{3.8}$$

One may picture $\cosh\omega$ and $\sinh\omega$ by using the unit hyperbola $x^2 - t^2 = 1$ in exactly the way ordinary sine and cosine are pictured on the unit circle.

Substituting these variables into the flat Minkowski metric gives

$$ds^2 = -\rho^2 d\omega^2 + d\rho^2. \tag{3.9}$$

This is still flat spacetime. Nothing physical has changed. Only the coordinates are different. Susskind explicitly tells the class to remember this formula and even leaves it on the board, because it is the form he wants to recognize later in the near-horizon Schwarzschild geometry.

A useful difference from ordinary polar coordinates is that θ is periodic, while ω is not. The variable ω runs from $-\infty$ to $+\infty$, and as $|\omega|$ becomes large the hyperbola approaches the light cone. In this sense ω behaves like a time variable adapted to uniformly accelerated observers.

3.4 The Near-Horizon Reduction

Susskind now returns to the black hole and states the real target of the calculation: the Schwarzschild metric near $r = 2MG$ should reduce to the same hyperbolic-polar form just written for flat space.

The first step is only algebra. Rewrite the radial-time part of the metric as

$$ds^2 = -\left(1 - \frac{2MG}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2MG}{r}} \tag{3.10}$$

$$= -\frac{r - 2MG}{r}\, dt^2 + \frac{r}{r - 2MG}\, dr^2. \tag{3.11}$$

Nothing has been approximated yet.

The approximation enters only after Susskind says that he wants to zoom in on $r = 2MG$. There is one delicate point, and he slows down to emphasize it. Near the horizon, one may replace r by $2MG$ where it varies slowly, but one should *not* tamper with $r - 2MG$, because that quantity changes sign across the horizon. With that rule, the metric becomes

$$ds^2 \approx -\frac{r - 2MG}{2MG}\, dt^2 + \frac{2MG}{r - 2MG}\, dr^2. \tag{3.12}$$

This approximation is not exact, but it becomes arbitrarily accurate as one restricts attention more and more tightly to the neighborhood of $r = 2MG$.

The next move is a change of radial coordinate. Susskind chooses ρ so that the entire radial term simply becomes $d\rho^2$:

$$d\rho^2 = \frac{2MG}{r - 2MG}\, dr^2. \tag{3.13}$$

Taking square roots gives

$$d\rho = \sqrt{\frac{2MG}{r - 2MG}}\, dr, \tag{3.14}$$

and integrating gives

$$\rho = 2\sqrt{2MG}\,\sqrt{r - 2MG}. \tag{3.15}$$

Squaring this relation produces the especially useful form

$$\rho^2 = 8MG(r - 2MG), \qquad r - 2MG = \frac{\rho^2}{8MG}. \tag{3.16}$$

Now substitute back into the time term. One finds the intermediate near-horizon metric

$$ds^2 = -\frac{\rho^2\, dt^2}{16M^2G^2} + d\rho^2. \tag{3.17}$$

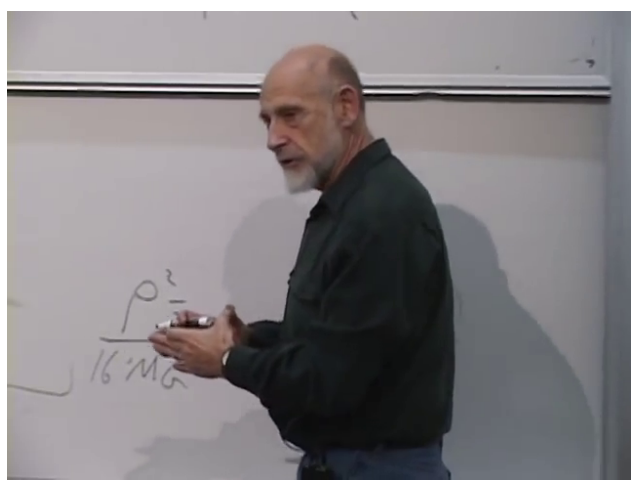


Figure 3.2: Board evidence for the intermediate near-horizon factor $\rho^2/(16M^2G^2)$. The full metric is reconstructed from the transcript; the frame secures only the key coefficient.

Only one nuisance factor remains. Rescale Schwarzschild time by

$$\omega = \frac{t}{4MG}, \qquad d\omega^2 = \frac{dt^2}{16M^2G^2}. \tag{3.18}$$

Then the metric becomes

$$ds^2 = -\rho^2 d\omega^2 + d\rho^2. \tag{3.19}$$

This is exactly the metric of flat Minkowski space in hyperbolic polar coordinates.

That is the central calculation of the lecture. Near $r = 2MG$, the Schwarzschild horizon is not a place where local forces blow up or where an infalling observer encounters a violent singularity. The apparently singular behavior of the original metric was a coordinate artifact. In Susskind's own rhythm, the horizon turns out to be a great big nothing.

Remark 3.2. A student asks what happens for $r < 2MG$, where the formula above would make ρ imaginary. Susskind does not try to develop the full interior extension here. His point in this lecture is narrower: the odd behavior at $r = 2MG$ is a coordinate glitch, not a physical catastrophe.

3.5 The Horizon as a Causal Boundary

With the algebra done, the lecture deliberately changes register. Susskind redraws the geometry and asks what these new coordinates mean physically. The key identification is immediate:

$$\rho = 0 \iff r = 2MG. \tag{3.20}$$

Thus the horizon is the place where the hyperbolic-polar coordinates pinch off. Outside the black hole, $r > 2MG$, so $\rho^2 > 0$, and the constant-ρ curves are the hyperbolae in the right Rindler wedge.

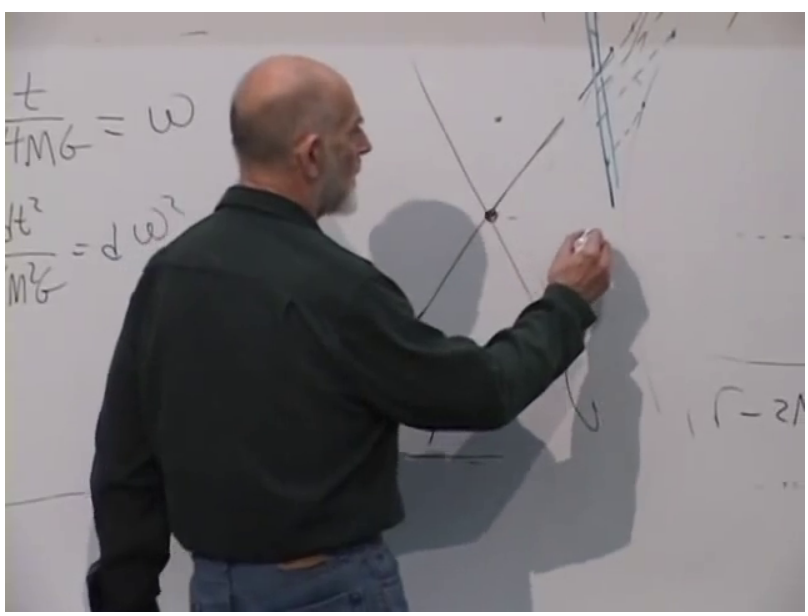


Figure 3.3: Board evidence for the near-horizon causal sketch, the ω-rescaling equations on the left, and the fixed-r construction on the right.

The equations written near that sketch can be cleaned up as

$$\omega = \frac{t}{4MG}, \tag{3.21}$$

$$d\omega^2 = \frac{dt^2}{16M^2G^2}, \tag{3.22}$$

$$ds^2 = -\rho^2 d\omega^2 + d\rho^2. \tag{3.23}$$

Along a fixed-ρ worldline one has $d\rho = 0$, so

$$d\tau = \rho\, d\omega. \tag{3.24}$$

Thus ω is not itself proper time, but on each constant-ρ trajectory it becomes proper time after multiplication by the corresponding radius ρ, exactly as θ becomes arc length after multiplication by the radius in ordinary polar coordinates.

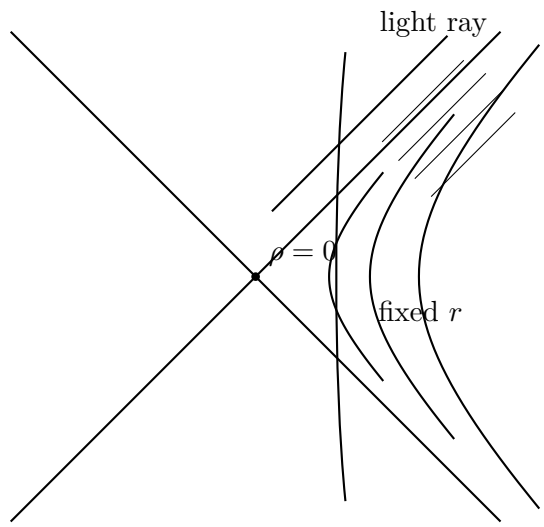


Figure 3.4: A restrained redraw of the same near-horizon picture: null boundaries, several fixed-ρ trajectories outside the horizon, the distinguished point $\rho = 0$, and the extra fixed-r construction suggested by the board sketch.

This is where Susskind brings in Bob and Alice. Bob insists on staying outside the black hole, so near the horizon he follows one of the constant-ρ hyperbolae. That means Bob is accelerating. Alice lets go and falls freely, so in the near-horizon flat-space picture she follows an inertial line and crosses the horizon without encountering anything locally special.

The asymmetry is entirely observational. Bob never sees Alice cross the horizon; he sees her approach it more and more slowly, asymptotically flattening against it. Alice, on the other hand, crosses with no local drama and can still see Bob for a while by looking back along her past light cone. Susskind repeatedly insists that this should not be turned into a metaphysical puzzle about what "really" happens. The profitable question is always: what can each observer measure, see, or signal?

A stationary outside observer also has a physical asymmetry relative to the infalling one. To remain at fixed r, Bob must accelerate, just as an observer standing on the ground accelerates relative to a freely falling frame. Nothing in the lecture suggests that Bob's description is wrong; rather, it is a description tied to accelerated worldlines that are causally barred from seeing through the horizon.

3.6 Acceleration Horizons

Susskind then steps back and says that none of this is uniquely about black holes. A uniformly accelerated observer in flat spacetime already has the same causal limitation. Such an observer sees a horizon beyond which signals never arrive and regards that boundary as the edge of the observable world. This is the acceleration horizon.

That remark is not a side comment. It is the conceptual payoff of the whole near-horizon analysis. The black-hole horizon, at least locally, is the gravitational version of the Rindler horizon of an accelerated observer. In this sense the lecture realizes Einstein's advice: to understand gravity, first understand acceleration.

3.7 Temperature, Negative Specific Heat, and Evaporation

After the break Susskind explicitly turns back to the previous lecture and puts on what he calls his "thermodynamic hat." The guiding picture is the same as before: build a black hole by dropping in photons, each carrying one bit of information. That picture already suggested a relation between area and entropy. Now he wants to extract temperature and then use it to estimate the evaporation rate.

Temperature as the energy cost of one bit. Susskind pauses to define temperature in the operational way he wants to use it. Temperature is not taken as a primitive sensory notion; it is the minimum energy change required to increase the entropy by one unit, or in the language of the lecture, by one bit. For the black hole, one adds a photon whose wavelength is comparable to the hole size,

$$\lambda \sim R_s \sim \frac{2GM}{c^2}. \tag{3.25}$$

The energy of one such photon is

$$E_\gamma \sim \frac{\hbar c}{\lambda}. \tag{3.26}$$

Therefore, up to the order-one factors that the heuristic argument does not yet control,

$$T \sim \frac{\hbar c}{\lambda} \sim \frac{\hbar c^3}{GM}. \tag{3.27}$$

This is the lecture's first-pass estimate: it is parametric, not exact.

Only after this scaling argument is on the board does Susskind quote the properly normalized Hawking temperature,

$$T_H = \frac{\hbar c^3}{8\pi GM}. \tag{3.28}$$

The important qualitative lesson is already visible:

$$T_H \propto \frac{1}{M}. \tag{3.29}$$

Large black holes are cold; small black holes are hot. The temperature is also manifestly quantum mechanical, since it vanishes with $\hbar$.

Negative specific heat. This inverse dependence on M is exactly what makes black holes thermodynamically strange. Adding energy to an ordinary body usually raises its temperature. Adding energy to a black hole increases its mass and therefore lowers its temperature. A black hole has negative specific heat.

For a stellar-mass black hole, the lecture gives the order-of-magnitude estimate

$$T_H \sim 10^{-8}\ \mathrm{K}. \tag{3.30}$$

That is much colder than the present cosmic microwave background temperature, about 3 K. A solar-mass black hole in today's universe is therefore not primarily an emitter; it is an absorber. It swallows surrounding radiation and grows colder as it grows more massive. If instead a black hole were hotter than its environment, it would radiate, lose mass, become hotter still, and run away in the opposite direction. Thermal equilibrium is unstable.

Susskind emphasizes that this instability is not peculiar to black holes alone. It is a familiar feature of gravitating systems more generally. A star that loses energy tends to contract and heat up, so gravity naturally produces systems with negative specific heat.

Evaporation in Planck units. To estimate the evaporation rate, Susskind now drops all the constants and works in Planck units,

$$c = \hbar = G = 1. \tag{3.31}$$

In these units energy is mass, so $E = M$. The luminosity of any warm body is written, up to order-one factors, as

$$L = -\frac{dE}{dt} \sim AT^4, \tag{3.32}$$

the Stefan–Boltzmann law. For a black hole,

$$A \sim R_s^2 \sim M^2, \qquad T \sim \frac{1}{R_s} \sim \frac{1}{M}. \tag{3.33}$$

Substituting these scalings gives

$$-\frac{dM}{dt} \sim AT^4 \sim M^2 \left(\frac{1}{M}\right)^4 = \frac{1}{M^2}. \tag{3.34}$$

Equivalently,

$$M^2 \frac{dM}{dt} \sim -1. \tag{3.35}$$

Up to the ignored factor of 3, this is the statement

$$\frac{d(M^3)}{dt} \sim -1, \tag{3.36}$$

so the evaporation time scales as

$$t_{\mathrm{evap}} \sim M^3. \tag{3.37}$$

This is the main quantitative result of the second half. It immediately explains why macroscopic black holes live for absurdly long times. The lecture gives the corresponding solar-mass estimate by converting to Planck units:

$$M_\odot \sim 10^{38} M_{\mathrm{Planck}} \qquad \Longrightarrow \qquad t_{\mathrm{evap}}(M_\odot) \sim 10^{114} t_{\mathrm{Planck}} \sim 10^{71}\ \mathrm{s} \sim 10^{64}\ \mathrm{yr}. \tag{3.38}$$

That is enormously longer than the present age of the universe. The later mountain-mass arithmetic in the lecture is too garbled to preserve confidently, but the scaling law $t_{\mathrm{evap}} \sim M^3$ and the qualitative lesson are completely clear: very cold, very large black holes evaporate extremely slowly.

What this points to. Susskind does not yet try to explain the detailed mechanism of Hawking emission. Instead he treats temperature, entropy, and evaporation as clues. A system with entropy behaves, in this respect, like a fluid: the smooth macroscopic description is not the whole story. Entropy is evidence for hidden microscopic degrees of freedom. In that spirit, black-hole thermodynamics is read not as a finished theory but as a strong hint that ordinary classical gravity is an effective description hiding a deeper microstructure.

The lecture ends by sharpening the tension rather than resolving it. If black holes radiate and eventually give back information, then evaporation must somehow be reconciled with the horizon picture in which matter falls through. Susskind frames this not as a solved paradox but as the central problem still to be understood.

3.8 Summary

The lecture makes two claims and then supports both of them carefully. First, the horizon of a large black hole is locally uneventful: after an appropriate change of coordinates, the near-horizon Schwarzschild metric becomes the flat Minkowski metric in hyperbolic polar form,

$$ds^2 = -\rho^2 d\omega^2 + d\rho^2.$$

Second, black holes are thermodynamic objects: they have a quantum temperature inversely proportional to their mass, a negative specific heat, and an evaporation time that scales like M^3 in Planck units.

What makes the lecture memorable is the order in which Susskind develops these points. He does not begin with polished final formulas. He begins with the apparent contradiction, detours into familiar coordinate systems, returns to the black hole only after the audience has the right model in mind, and then turns the resulting geometry into an operational story about observers, signals, and horizons. Only after the horizon has been made locally boring does he return to thermodynamics and let the real mystery reappear at a deeper level: if horizons are locally harmless but black holes have entropy and radiate, then gravity must be hiding more structure than the classical metric alone reveals.

Chapter 4

Black Holes, White Holes, and Penrose Diagrams

These notes follow Leonard Susskind's fourth lecture in the supplementary *Topics in String Theory* track of *Cosmology And Black Holes*, with curation by LazyingArt LLC. The lecture unfolds in two long arcs. In the first, Susskind returns to the near-horizon Schwarzschild geometry, asks how that local picture connects to the far region, and then uses observers, light rays, and the singularity to build the causal story of a black hole. In the second, after a short reset about proper distance, he introduces Penrose diagrams and uses them to separate the eternal Schwarzschild solution from the geometry of a black hole formed by collapse.

4.1 Near-Horizon Recap and Choice of Variables

Susskind opens by saying, in effect, let us go back to something familiar. Near the horizon the Schwarzschild geometry is approximately just flat spacetime, but only after one uses the right variables. The first is ρ, the proper distance from the horizon; the second is ω, a dimensionless time variable built from the Schwarzschild time t.

Remark 4.1. Throughout these notes the lecture's spoken m, g are standardized to M, G, $c = 1$ is understood, and the metric signs are written in a consistent Lorentzian convention. When the spoken transcript slips verbally, the displayed equations below follow the mathematically intended form.

The board recap is one of the clearest pieces of visual evidence in the lecture.

The two equations written explicitly on the board are

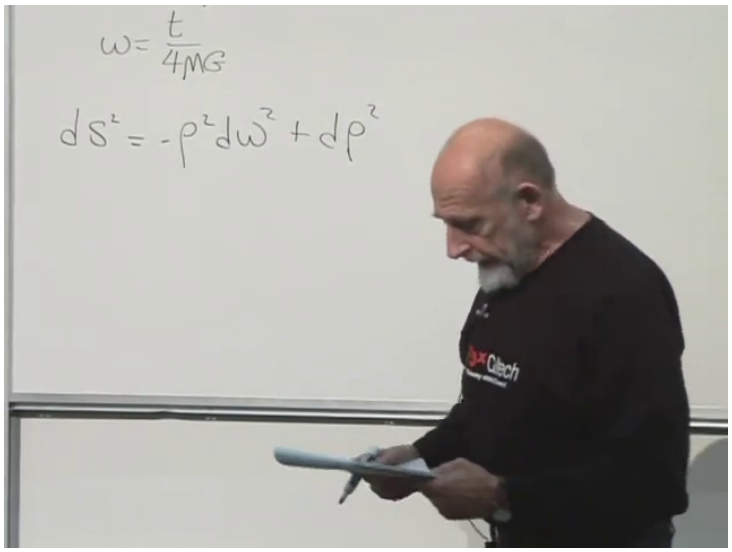
$$\omega = \frac{t}{4MG}, \tag{4.1}$$

$$ds^2 = -\rho^2\, d\omega^2 + d\rho^2. \tag{4.2}$$

Susskind also reminds the audience of the defining geometric fact

$$\rho = 0 \qquad \text{at} \qquad r = 2MG. \tag{4.3}$$

He does not re-derive the function $\rho(r)$, and for this lecture he does not need to. What matters is only that r is not the proper distance from the horizon, whereas ρ is.



The metric above is then compared to the metric of the Euclidean plane in polar coordinates,

$$ds^2 = \rho^2\, d\theta^2 + d\rho^2. \tag{4.4}$$

The comparison is the point. Near the horizon, the radial-time part of Schwarzschild is flat spacetime written in a hyperbolic-polar form, and ω plays the role that θ plays in ordinary polar coordinates. The only visible difference is the minus sign attached to the Lorentzian time direction.

This is not yet the whole black hole. It is the recalled local geometry near $r = 2MG$. But it immediately organizes the first picture: ω runs from the remote past to the remote future, the lines of constant ρ are hyperbolae, and the horizon itself lies at $\rho = 0$. With that local picture back on the board, Susskind can ask the first genuinely new question of the lecture: what happens far from the black hole, and what sort of geometry interpolates between the two regimes?

4.2 From the Horizon to Infinity: Why a Cigar Appears

With the near-horizon metric re-established, Susskind turns to the opposite limit. Very far from the black hole, one should again recover flat spacetime. The issue is not whether the geometry becomes flat, but how the coefficient of the time term changes between the horizon and infinity.

Suppressing the angular part, the relevant radial-time sector of the Schwarzschild geometry is

$$ds^2 = -\left(1 - \frac{2MG}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2MG}{r}}. \tag{4.5}$$

The transcript is verbally messy at this point, but the intended algebra is straightforward.

Asymptotic reduction.

1. Take $r \gg 2MG$, so that $1 - 2MG/r \to 1$.
2. In that region the Schwarzschild coordinate r and the proper distance ρ are approximately the same:

$$r \approx \rho.$$

3. The metric therefore reduces to

$$ds^2 \approx -dt^2 + dr^2 \approx -dt^2 + d\rho^2. \tag{4.6}$$

4. Using $t = 4MG\,\omega$, one has

$$dt^2 = 16M^2G^2\,d\omega^2. \tag{4.7}$$

5. Hence far from the black hole,

$$ds^2 \approx -16M^2G^2\,d\omega^2 + d\rho^2. \tag{4.8}$$

This is the comparison that drives the rest of the section:

$$ds^2 \approx -\rho^2\,d\omega^2 + d\rho^2 \qquad \text{near the horizon,} \tag{4.9}$$

$$ds^2 \approx -16M^2G^2\,d\omega^2 + d\rho^2 \qquad \text{far away.} \tag{4.10}$$

The radial piece is not the interesting part. The interesting part is that the coefficient of $d\omega^2$ changes from ρ^2 near the horizon to a constant far away.

This also sharpens Susskind's remarks about clocks. Along a hovering worldline with $d\rho = 0$, the near-horizon metric gives

$$d\tau^2 = \rho^2\,d\omega^2, \tag{4.11}$$

so very close to the horizon only a very small proper time elapses for a given increment of ω. Far away, by contrast,

$$d\tau^2 = 16M^2G^2\,d\omega^2, \tag{4.12}$$

so the same ω-interval corresponds to the ordinary asymptotic time scale $dt = 4MG\,d\omega$.

At this point Susskind deliberately changes register. His own comment is that it is much easier to understand space than spacetime, so he introduces an ordinary Euclidean model with the same radial dependence in the angular coefficient:

$$ds^2 \sim \rho^2\,d\theta^2 + d\rho^2 \qquad \text{near the tip,} \tag{4.13}$$

and

$$ds^2 \sim 16M^2G^2\,d\theta^2 + d\rho^2 \qquad \text{far away.} \tag{4.14}$$

This auxiliary geometry is not the black-hole spacetime itself. It is a visual aid for the changing coefficient.

The answer is the "cigar" geometry. Near the tip, the surface looks flat, just like ordinary polar coordinates near the origin. Far away, the distance around the θ-direction no longer grows with ρ; it approaches a constant circumference. Since θ runs from 0 to 2π, that asymptotic circumference is

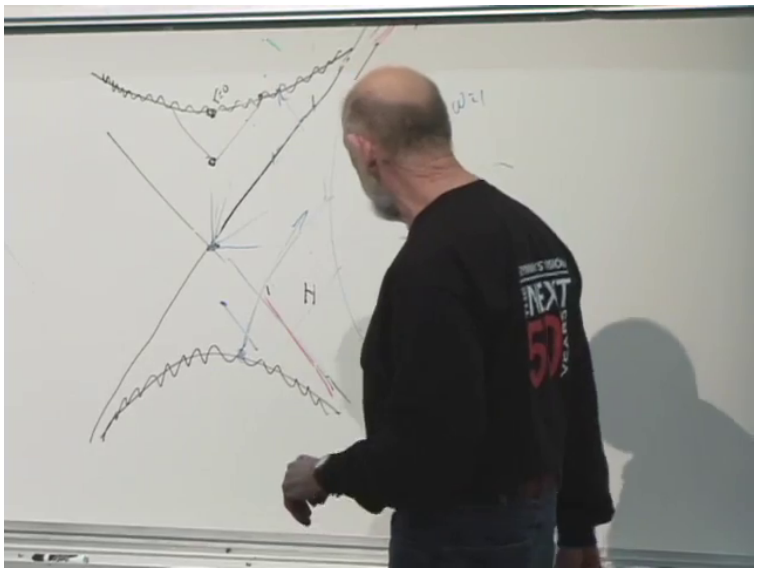
$$\oint ds = \int_0^{2\pi} 4MG\,d\theta = 8\pi MG. \tag{4.15}$$

The geometric lesson is immediate. A large black hole corresponds to a cigar with a large asymptotic girth and therefore a blunter, less curved tip. A small black hole corresponds to a tighter cigar and larger curvature near the tip. Susskind does not introduce a curvature scalar here; the point is the picture.

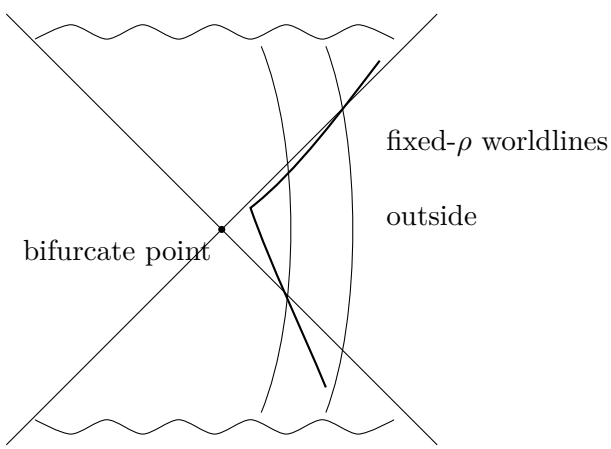
He then returns to the main statement. The black-hole geometry is the one for which the coefficient of $d\omega^2$ behaves like ρ^2 near the horizon and like $16M^2G^2$ far away. The cigar is the clearest ordinary-space analogue of that interpolation.

4.3 Horizon Geometry and the Causal Sketch

Once the interpolation has been made visual, Susskind returns to the Lorentzian spacetime picture. Here the lecture changes emphasis: from metric coefficients to causal meaning. The relevant board sketch is developed rather than pristine, but that is part of its value, since it preserves the way the argument accumulated on the blackboard.



A cleaned version of the same structure is shown below. It is not meant as a literal transcription of every chalk stroke. It isolates the features that the transcript makes secure: crossed null lines, the central bifurcation point, fixed-ρ worldlines outside the horizon, and the wavy future and past boundaries that will later be identified as singularities.



The rule for reading all such sketches is one that Susskind repeats several times during the lecture.

Definition 4.2. In the reduced radial spacetime diagrams used here, radial light rays are drawn at 45°.

With that rule in hand, Susskind distinguishes two notions of horizon. The first is the *bifurcate horizon*, which in his terminology is the central crossing point of the two null lines. The second, and

physically more important one, is the *event horizon*: the entire null surface that acts as the boundary of communication. Once this distinction is made, the sketch is no longer just a diagram. It becomes a statement about signaling. Light rays can fall inward through the horizon, but outward-directed radial light rays emitted from the interior do not escape to the exterior.

This is also where the fixed-ρ curves become physical. They represent observers hovering outside the black hole at constant proper distance from the horizon. In the lecture these worldlines are not yet Bob's story in full; they are first the geometric realization of what it means to remain outside the hole.

4.4 Bob, Alice, Redshift, and the One-Way Barrier

At this point the lecture becomes operational. Susskind moves from geometry to what particular observers can see, measure, and signal. Bob remains outside on a fixed-ρ worldline. Alice falls inward. The diagram is now read not merely as geometry but as an account of communication.

For Bob, the time parameter ω is just the Schwarzschild time in units of $4MG$:

$$t = 4MG\,\omega. \tag{4.16}$$

Thus the marks $\omega = 0, 1, 2, \ldots$ along Bob's worldline are simply $t = 0, 4MG, 8MG, \ldots$. He stays outside; he does not cross the horizon; he can keep reading off more and more time on his wristwatch.

Susskind first phrases the issue in terms of a photon, then quickly turns it into the more useful story of Alice, because Bob can only *see* Alice by receiving light emitted from her. The causal sequence is then the following. Bob looks back and receives light from progressively later points of Alice's infalling worldline, but each received signal comes from a point still outside the horizon. He keeps seeing her wave. He keeps waiting. He keeps receiving signals. Yet the signals arrive farther and farther apart in his own proper time.

The crucial observation is that Bob never sees Alice cross the horizon. In the classical picture she appears to slow down, freeze just outside the horizon, and remain there. Susskind immediately refines that frozen picture into a statement about frequency. Alice is visible only because charges in her body are oscillating and emitting electromagnetic radiation. If Bob sees those oscillations slow down, then the light he receives must be redshifted. The lecture follows that redshift concretely:

- first optical light,
- then infrared,
- then long-wavelength radio,
- and finally wavelengths so long that Bob effectively sees nothing at all.

The practical content of "freezing at the horizon" is therefore not that Alice remains optically visible forever, but that she fades away.

Susskind then asks for the reverse point of view. What does Alice see? Here the lecture is careful, because one can easily overread Bob's story and imagine that something dramatic must happen exactly at the horizon. Alice's experience says otherwise. She continues to see Bob even after she has crossed the horizon. Bob can still send signals inward to her. She, however, cannot send signals back out once she is inside. The horizon is therefore a one-way barrier, but not in the naive sense

that each observer sees the same thing reversed. Bob sees Alice redshift and disappear; Alice sees Bob recede and redshift, but she does not see him vanish at her moment of crossing.

The lecture then sharpens the point with the feet-first thought experiment. Alice falls feet first, so the worldline of her feet crosses the horizon before the worldline of her head. If she simply continues to fall, nothing special happens at the horizon. She can still look back and see light coming from her feet, just as from any nearby object: not from the simultaneous event on the foot worldline, but from a slightly earlier point on it. In that sense there is no paradox. Horizon crossing by itself does not locally rip the body apart.

The trouble begins only if Alice changes her mind. Suppose her feet have crossed the horizon but her head tries to remain outside. Then her head must accelerate away from the black hole. The part already inside cannot accelerate enough to keep up, and in Susskind's description the body is stretched farther and farther until it breaks. This is not a generic statement that the horizon is locally singular. It is a more specific one: once part of an extended body has crossed the one-way barrier, trying to keep the rest of the body out requires an impossible kinematic arrangement.

4.5 The Singularity, Coordinate Interchange, and White Holes

Only after the horizon story has been made vivid does Susskind return to the Schwarzschild metric and asks where the singularity lies on the diagram. The order matters. He first separates the horizon from the singularity, and only then turns to the singularity itself.

The radial-time Schwarzschild sector is again

$$ds^2 = -\left(1 - \frac{2MG}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2MG}{r}}. \tag{4.17}$$

The immediate fact is the sign change:

$$1 - \frac{2MG}{r} > 0 \quad \text{for} \quad r > 2MG, \qquad 1 - \frac{2MG}{r} < 0 \quad \text{for} \quad r < 2MG. \tag{4.18}$$

Outside the horizon the coefficient of dt^2 is negative and the coefficient of dr^2 is positive, so t is timelike and r spacelike. Inside the horizon the signs reverse. Susskind's point is that this does *not* mean that the horizon is physically singular. It means that Schwarzschild coordinates become peculiar there: the coordinate roles interchange.

The lecture then gives this coordinate fact a geometric reading. One may imagine moving inward along a line of constant Schwarzschild time until one reaches $r = 2MG$. What happens next is not that one arrives at a special material surface. Rather, in these coordinates the r-lines turn and continue in a timelike direction. The important destination is not $r = 2MG$, but

$$r = 0. \tag{4.19}$$

That is the singularity. There tidal forces diverge, curvature becomes infinite, and, in Susskind's phrasing, all hell breaks loose.

The lecture insists on a further conceptual correction. At first one is tempted to think of the singularity as a *place*. But on the causal sketch it is not drawn as a vertical worldline. It is drawn as a spacelike hyperbola-like boundary in the future. In Susskind's own language, it is more like a *time* than a place: once one has crossed the horizon, the singularity lies ahead in one's future. This

is why the black-hole interior is not merely a region from which one cannot communicate outward. It is a region in which every future-directed timelike or null path ends at the singularity.

From here the lecture takes its next conceptual jump. Schwarzschild is time-independent, and dt^2 is unchanged under $t \to -t$. The geometry is therefore time-reversal invariant. If a history in which Alice falls through a future horizon into a future singularity is allowed, then the time-reversed history must also solve the equations. That reversed history contains a past singularity and a past horizon from which matter may emerge. This is the *white hole*: the time reverse of the black hole.

Susskind is quite explicit about the contrast between mathematics and plausibility. Mathematically, if Alice can pass in through a future horizon, then in the reversed solution she can pass out through a past horizon. Physically, however, that story looks absurd: Bob would see Alice suddenly ejected from a singular past. The lecture does not yet resolve that absurdity; it merely puts it on the table.

4.6 Entropy and Why the White-Hole Half Is Unphysical

The next section of the lecture is not a new geometric construction. It is a sustained explanation of why the time-reversed half of the eternal Schwarzschild solution is mathematically present but physically fantastical. Susskind does not treat this as a contradiction in the equations. He treats it as an entropy problem.

The structure of the argument is the familiar structure of irreversible physics. A bomb explodes and scatters debris; the time reverse, in which debris flies inward and delicately reassembles the bomb, is not forbidden by Newton's equations, but it is overwhelmingly improbable. A ball rolls down a hill; the reverse trajectory, in which it comes to rest exactly on the top, is possible but exquisitely unstable. Fifteen billiard balls scatter from a triangular rack; their spontaneous reconvergence into a perfect triangle is possible in principle and ridiculous in practice. An ice cube melts in a pot of hot water; the reverse process, in which an ice cube assembles itself from thermal motion, is again allowed by the microscopic equations and fantastically unlikely.

Susskind's point is that the white hole belongs in exactly this class. It is not *impossible* in the formal time-reversal sense. It is the analogue of the entropy-lowering reverse trajectory. What Bob would see in the white-hole picture is not a forbidden process, but an outrageously special one: the black hole suddenly ejecting a full-fledged Alice. The lecture's language varies between "nonsense," "fiction," and "unlikely," but the conceptual content is consistent. The reason to distrust the white-hole half is not that the equations ban it. The reason is that it is an enormous entropy-decreasing fluctuation.

This is also where the lecture connects the discussion back to black-hole thermodynamics. Black holes have entropy. Their horizons encode hidden information. Rare thermal or quantum fluctuations can in principle do strange things, but they do not characterize ordinary black-hole physics. In that sense the white-hole half of the eternal diagram is kept as a mathematical artifact and discarded as a model of real astrophysical black holes.

4.7 Penrose Diagrams: First Flat Space, Then Schwarzschild

After the break Susskind deliberately resets the audience. He notices that he has used the phrase "proper distance" as jargon and stops to explain it. The transcript is garbled at points, but the

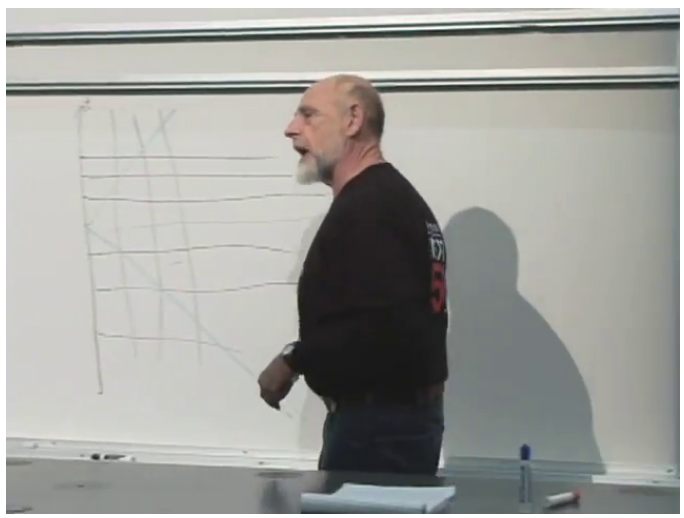
secure content is clear.

Definition 4.3. Proper distance is distance measured by honest meter sticks along the chosen spatial path.

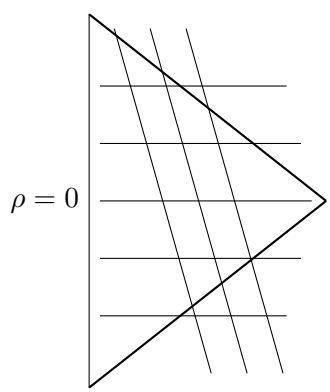
That clarification prepares the transition to Penrose diagrams. Susskind's attitude here is pragmatic. One does not need the detailed coordinate formulas in order to use such diagrams. One needs only the reading rule.

Definition 4.4. A Penrose diagram is a compactified spacetime diagram in which an infinite spacetime region is mapped to a finite domain while radial light rays remain at 45°.

He begins with flat spacetime, not with the black hole. The ordinary (t, ρ) half-plane has $\rho \geq 0$, a vertical time axis, a horizontal radial axis, and an infinite grid of constant-t and constant-ρ lines. The construction-stage board image is preserved below.



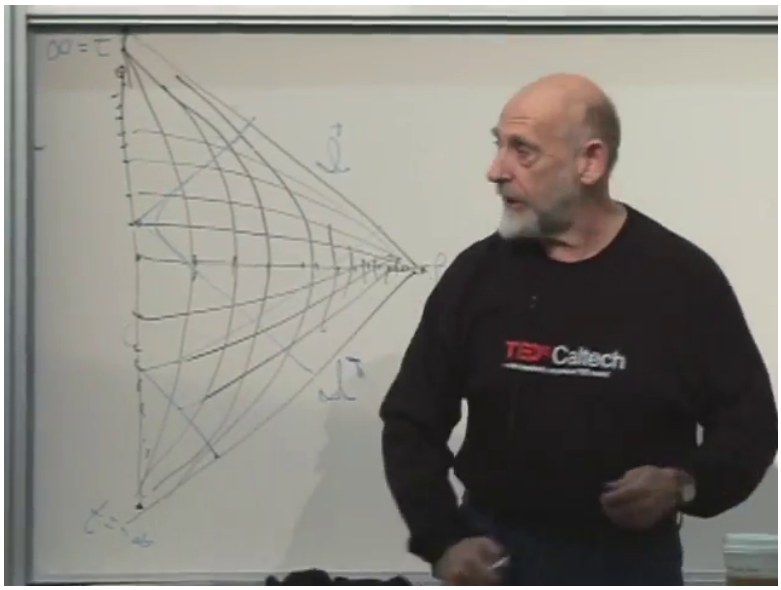
A minimal reconstruction of that intermediate step is:

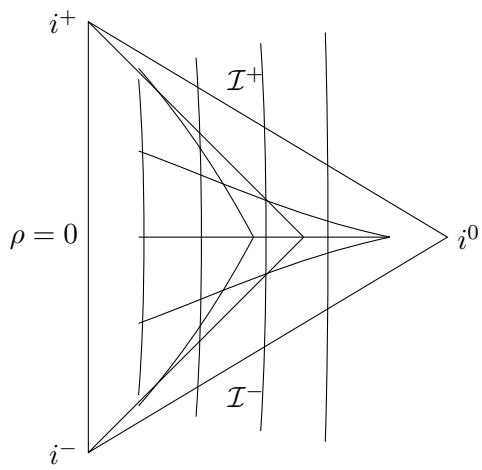


This is not yet the final Penrose diagram. It is the rule being enacted on the board: squash the infinite t-ρ grid into a finite region while preserving the 45° direction of radial light.

The more complete board result is then:

A clean version of the compactified flat-space Penrose diagram is:





The notation here is the standard cleaned notation for Susskind's spoken labels:

$$i^+, \quad i^-, \quad i^0, \quad \mathcal{I}^+, \quad \mathcal{I}^-. \tag{4.20}$$

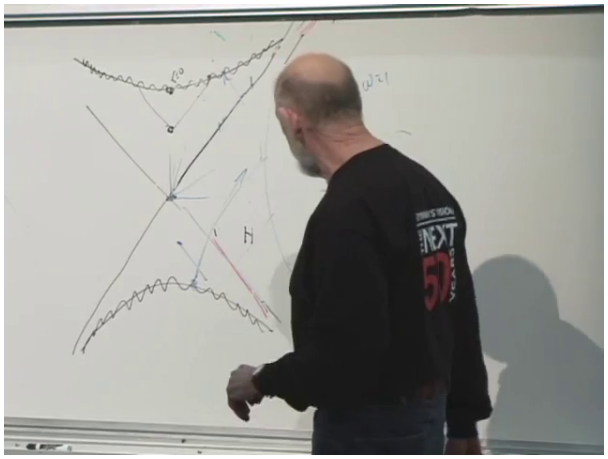
The left boundary is the timelike axis $\rho = 0$. The right tip is spacelike infinity i^0. The slanted upper and lower boundaries are future and past null infinity, the "scri plus" and "scri minus" of the lecture.

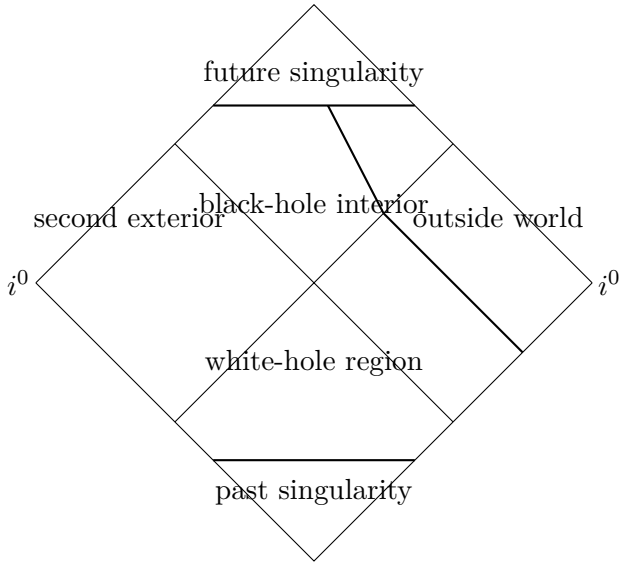
Remark 4.5. As Susskind later emphasizes in response to a question, the 45° rule is a rule for radial light rays. A light ray that misses the origin is not purely radial in the reduced two-dimensional description and need not be represented by a straight 45° line throughout.

Only after flat space has been taught as a reading exercise does Susskind turn back to the black hole. The causal sketch from earlier in the lecture is the pre-compactification object to be "squished."

A standard cautious reconstruction of the eternal Schwarzschild Penrose diagram is then:

The diagram collects in one place much of what the lecture has already established locally. Radial null rays still move at 45°. An inward-directed light ray from the exterior crosses the horizon and





terminates on the future singularity. The upper triangular region is the black-hole interior. The lower one is the white-hole region. The right-hand diamond is the exterior world in which Bob lives. The left-hand diamond is a second asymptotic exterior region present in the eternal Schwarzschild solution.

Susskind's next move is to undercut the physical importance of half this picture. The second exterior and the white-hole half are artifacts of an eternal solution. They are not the geometry of an ordinary black hole formed in nature.

4.8 Real Black Holes and the Birkhoff-Theorem Preview

The lecture closes by shifting from eternal black holes to real ones. In nature, black holes are formed. Susskind imagines matter, even a cloud of photons, coming in from far away. Since photons carry energy and therefore gravitate, enough incoming radiation concentrated in a sufficiently small region can form a black hole. The question is then not what an eternal Schwarzschild solution looks like, but how one glues a nearly empty pre-collapse spacetime to a black-hole spacetime in the future.

To prepare that construction, Susskind introduces first the Newtonian theorem and then its general-relativistic analogue.

Proposition 4.6 (Newton's shell theorem)**.** *For a spherically symmetric shell of total mass M, the Newtonian gravitational field vanishes in the interior of the shell, while in the exterior it is exactly the same as the field of a point mass M placed at the center.*

Susskind phrases this in the language of lines of force or flow lines of a fluid. Inside the shell there is no place for the symmetric inward flow to go, so the field must vanish. Outside the shell the lines of force are exactly those of a point mass at the center. He also emphasizes that this remains true if the shell changes radius in time, provided the total mass is conserved.

The general-relativistic statement is Birkhoff's theorem in the operational form needed for the next lecture.

Theorem 4.7 (Birkhoff's theorem in the form used in the lecture)**.** *For a spherically symmetric shell of total mass M, the spacetime inside the shell is flat Minkowski spacetime, while the spacetime outside the shell is the Schwarzschild geometry of mass M.*

The two pieces to be patched together are therefore

$$ds^2_{\text{inside}} = -dt^2 + dx^2 + dy^2 + dz^2, \tag{4.21}$$

$$ds^2_{\text{outside}} = \text{Schwarzschild}(M). \tag{4.22}$$

As the shell contracts, its radius changes but its total mass does not. The boundary between the flat interior and the Schwarzschild exterior moves with time. That moving interface is exactly the structure Susskind wants for the next lecture, where the eternal diagram will finally be replaced by the spacetime of an actually formed black hole.

4.9 Summary

The lecture begins by returning to the near-horizon geometry in the variables ρ and ω, so that the local black-hole metric can be written as

$$ds^2 = -\rho^2\, d\omega^2 + d\rho^2.$$

From there Susskind asks how this local form connects to the far region, and he answers that question first by a cleaned asymptotic comparison and then by the Euclidean cigar analogy. With that intuition in hand, he turns the metric into a causal story: horizons as one-way barriers, Bob outside, Alice falling in, redshift, fading signals, and the feet-first thought experiment.

Only after that long operational discussion does he introduce the singularity and the coordinate interchange across $r = 2MG$, and from there the lecture naturally reaches the time-reversed white-hole extension. The white hole is left mathematically possible but physically downgraded by entropy. After the break, Penrose diagrams compress the same causal content into finite pictures, first for flat spacetime and then for eternal Schwarzschild. The chapter ends where the lecture ends: eternal black holes are not the real story, and Birkhoff's theorem provides the bridge to the next lecture's construction of black-hole formation by collapse.